MATH1025B
Essential Mathematical Methods

Chapter 11: Conic Sections

November 27, 2025

Conic Sections

Conic sections formed by cutting a double cone with a plane:

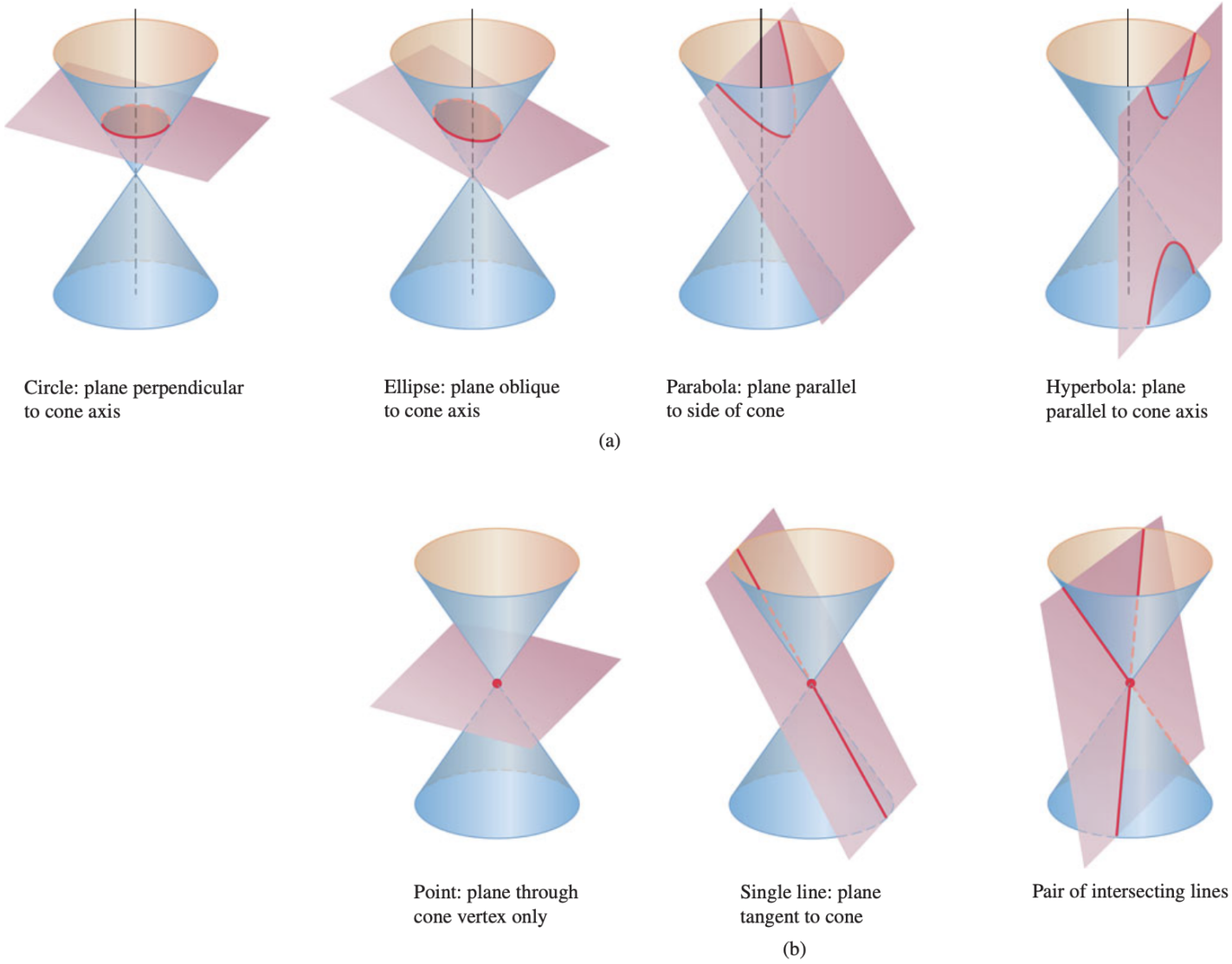


FIGURE 11.37 The standard conic sections (a) are the curves in which a plane cuts a *double* cone. Hyperbolas come in two parts, called *branches*. The point and lines obtained by passing the plane through the cone's vertex (b) are *degenerate* conic sections.

Definition 11.1

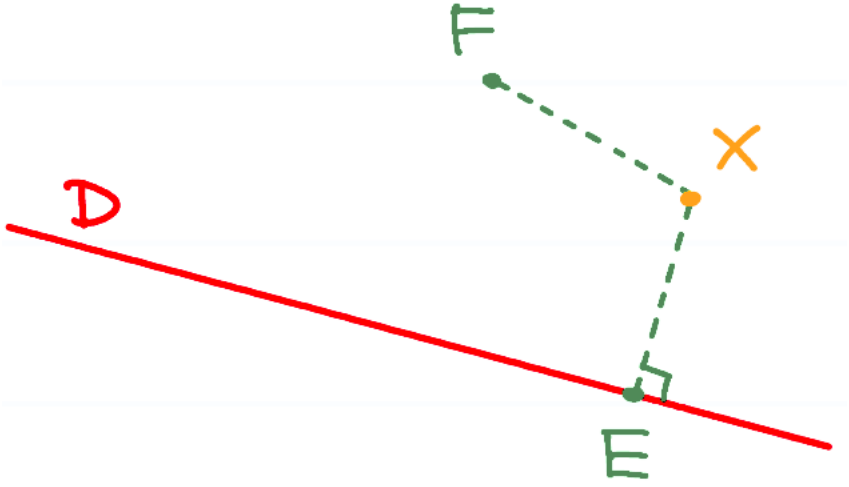
A **conic section** is the locus of a point X which moves in a plane in such a way that

the distance XF of X from a fixed point F (called **focus**) is a constant multiple of its distance from a fixed line D (called **directrix**)

i.e.

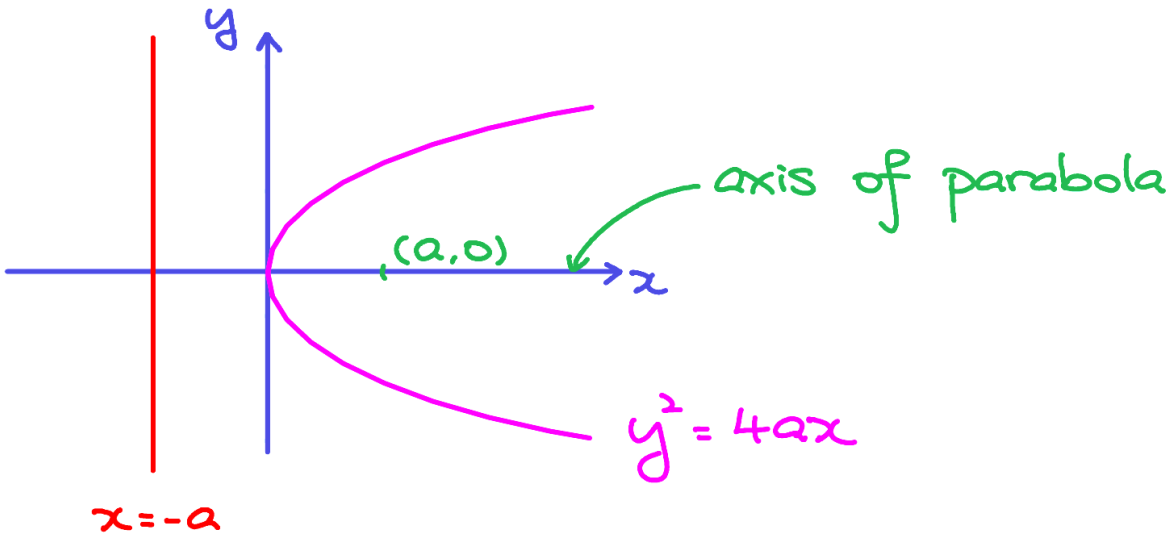
$$\frac{XF}{XE} = e,$$

where $e > 0$ is a constant (called **eccentricity**), and E is the foot of perpendicular from X to the directrix D.



Parabola ($e = 1$)

Parabola:
Assume $F = (a, 0)$ and $D : x = -a$, where $a > 0$.



Let $X = (x, y)$. Then

$$\begin{aligned} XF &= XE \\ \sqrt{(x-a)^2 + (y-0)^2} &= |x - (-a)| \\ (x-a)^2 + y^2 &= (x+a)^2 \\ y^2 &= 4ax. \end{aligned}$$

Parabola ($e = 1$)

Let $P(x_0, y_0)$ be a point on the parabola $\mathcal{P} : y^2 = 4ax$.
Differentiate both sides with respect to x:

$$2y\frac{dy}{dx} = 4a$$

$$\left.\frac{dy}{dx}\right|_{(x,y)=(x_0,y_0)} = \frac{2a}{y_0} \qquad (\text{if } y_0 \neq 0)$$

Tangent of $\mathcal{P}$ at $P(x_0, y_0)$:

$$\frac{y - y_0}{x - x_0} = \frac{2a}{y_0}$$

$$2ax - y_0 y + y_0^2 - 2ax_0 = 0$$

$$2ax - y_0 y + 2ax_0 = 0 \quad (\because y_0^2 = 4ax_0)$$

Normal of $\mathcal{P}$ at $P(x_0, y_0)$:

$$\frac{y - y_0}{x - x_0} = -\left(\frac{2a}{y_0}\right)^{-1} = -\frac{y_0}{2a}$$

$$y_0 x + 2ay - (2a + x_0)y_0 = 0$$

Parabola ($e = 1$)

Parametrization:
The curve $\gamma : \mathbb{R} \to \mathbb{R}^2$ defined by $\gamma(t) = (at^2, 2at)$ gives a parametrization of the parabola $\mathcal{P} : y^2 = 4ax$.
Let $P(at_0^2, 2at_0)$ be a point on the parabola $\mathcal{P} : y^2 = 4ax$.

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t} \quad (\text{if } t \neq 0) \implies \frac{dy}{dx}\Big|_{t=t_0} = \frac{1}{t_0}$$

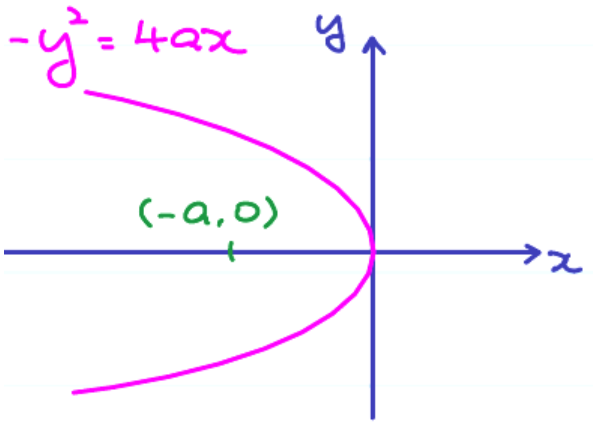
Tangent of $\mathcal{P}$ at $P(at_0^2, 2at_0)$:

$$\frac{y - 2at_0}{x - at_0^2} = \frac{1}{t_0}$$
$$x - t_0 y + at_0^2 = 0$$

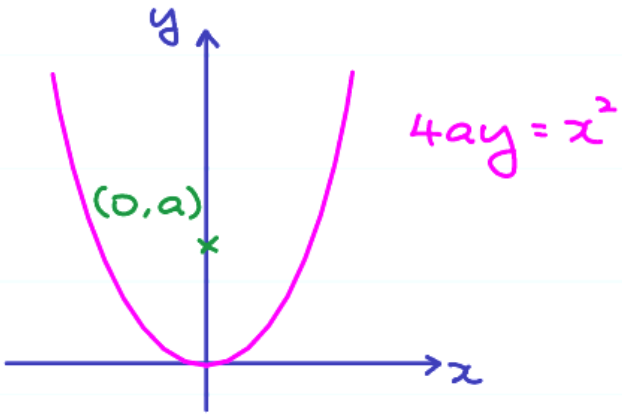
Normal of $\mathcal{P}$ at $P(at_0^2, 2at_0)$:

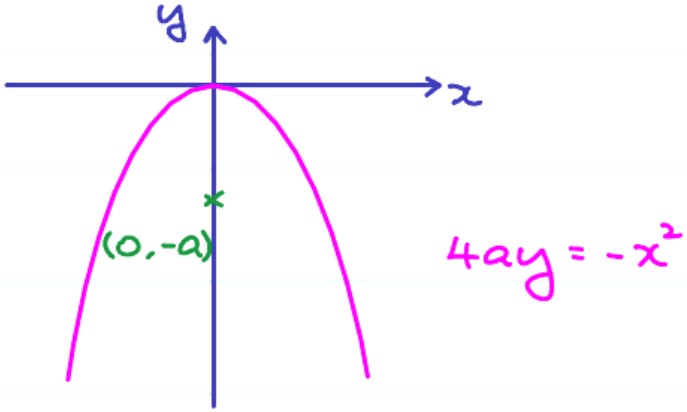
$$\frac{y - 2at_0}{x - at_0^2} = -t_0$$
$$t_0 x + y - 2at_0 - at_0^3 = 0$$

Other forms:
Replace a by $-a$:



Interchange the roles of x and y:





Example 11.1

How many tangents of the parabola $\mathcal{P} : y^2 = 4ax$ pass through $Q = (h, k)$?

Let $L : x - ty + at^2 = 0$ be a tangent of $\mathcal{P}$ passing through $Q(h, k)$. Then

$$at^2 - kt + h = 0.$$

The equation has

- two solutions (i.e. $\mathcal{P}$ has two tangents at (h, k)) if
$$k^2 - 4ah > 0, \quad \text{i.e. } (h, k) \text{ is on the left hand side of } \mathcal{P}$$
- no solution (i.e. $\mathcal{P}$ has no tangents at (h, k)) if
$$k^2 - 4ah < 0, \quad \text{i.e. } (h, k) \text{ is on the right hand side of } \mathcal{P}$$
- exactly one solution (i.e. $\mathcal{P}$ has exactly one tangent at (h, k)) if
$$k^2 - 4ah = 0, \quad \text{i.e. } (h, k) \text{ lies on } \mathcal{P}$$

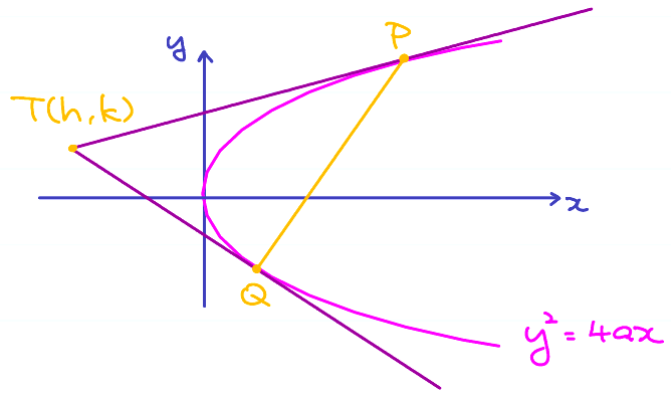
Exercise 11.1

Find the two tangents of $y^2 = 4x$ passing through $(-2, 1)$.

Ans: $x - 2y + 4 = 0$ and $x + y + 1 = 0$

Example 11.2

Let $T(h, k)$ be an external point of $\mathcal{P} : y^2 = 4ax$. Find the equation of chord PQ, where P and Q are the points of contact of the tangents from T.



Let $P(x_1, y_1)$, $Q(x_2, y_2)$.

$$\text{Equation of tangent at } P : 2ax - y_1 y + 2ax_1 = 0$$

$$\text{Equation of tangent at } Q : 2ax - y_2 y + 2ax_2 = 0$$

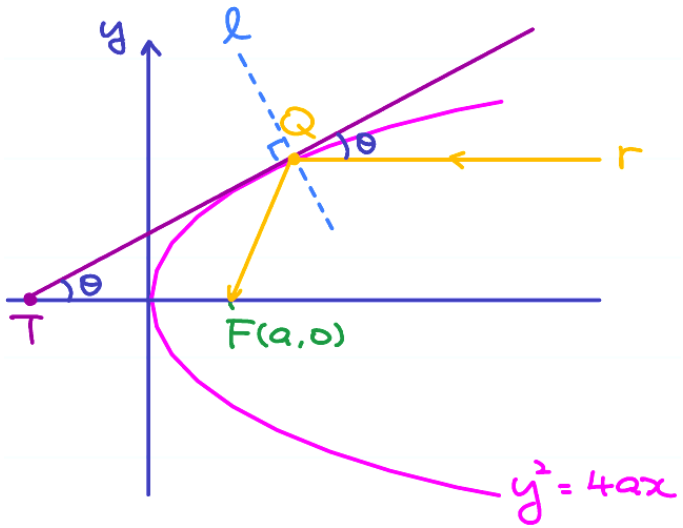
Since T lies on both tangents, we have

$$2ah - y_1 k + 2ax_1 = 0 \quad \text{and} \quad 2ah - y_2 k + 2ax_2 = 0.$$

Consider the line $2ah - yk + 2ax = 0$.
Since both $P(x_1, y_1)$, $Q(x_2, y_2)$ satisfy $2ah - yk + 2ax = 0$, it is the equation of PQ.

Example 11.3 (Parallel rays to axis of parabola will converge to focus)

Let $Q = (at^2, 2at)$ be a point on $\mathcal{P} : y^2 = 4ax$. Then the normal ℓ of $\mathcal{P}$ at Q bisects the ray r and FQ. Equivalently, $\angle FQT = \angle FTQ$.



Suppose the tangent at Q cuts the x-axis at an angle θ.

$$\text{Slope of } QT = \frac{1}{t} = \tan\theta$$

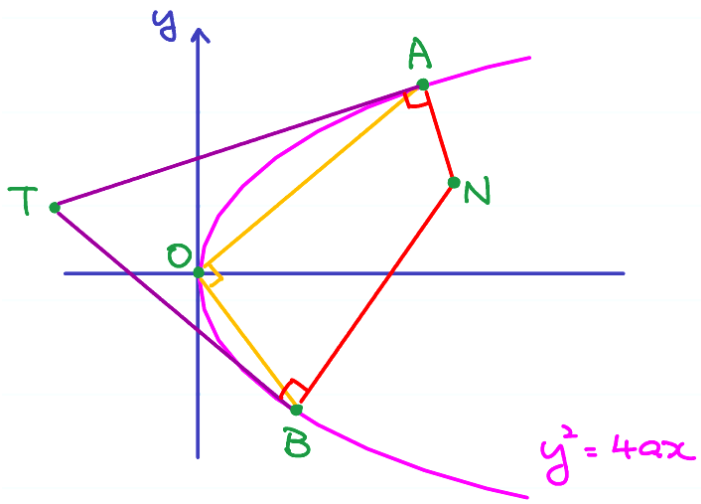
$$\text{Slope of } QF = \frac{2at - 0}{at^2 - a} = \frac{2t}{t^2 - 1} = \frac{2(\frac{1}{t})}{1 - (\frac{1}{t})^2} = \frac{2\tan\theta}{1 - \tan^2\theta} = \tan 2\theta$$

So $\angle FQT = 2\theta - \theta = \theta = \angle FTQ$.

Example 11.4

A and B are points on the parabola $y^2 = 4ax$ moving in a way that $\angle AOB$ is a right angle. Suppose that TA and TB are tangents, NA and NB are normals. Show that the locus of the mid-point M of TN is a parabola.

Let $A = (at_1^2, 2at_1)$, $B = (at_2^2, 2at_2)$. Then



$$\text{Eq. of } TA : x - t_1 y + at_1^2 = 0 \qquad (1)$$

$$\text{Eq. of } TB : x - t_2 y + at_2^2 = 0 \qquad (2)$$

Solving (1) and (2): $T = (at_1t_2, a(t_1 + t_2))$.

$$\text{Eq. of } NA : t_1 x + y - 2at_1 - at_1^3 = 0 \qquad (3)$$

$$\text{Eq. of } NB : t_2 x + y - 2at_2 - at_2^3 = 0 \qquad (4)$$

Solving (3) and (4): $N = (a(2 + t_1^2 + t_1t_2 + t_2^2), -at_1t_2(t_1 + t_2))$.

$$\therefore M = \frac{1}{2}(T + N) = \left(\frac{a}{2}\left[2 + (t_1 + t_2)^2\right], \frac{a}{2}(t_1 + t_2)(1 - t_1t_2)\right)$$

Example 11.4

A and B are points on the parabola $y^2 = 4ax$ moving in a way that $\angle AOB$ is a right angle. Suppose that TA and TB are tangents, NA and NB are normals. Show that the locus of the mid-point M of TN is a parabola.

Set

$$(x, y) = M = \left(\frac{a}{2}\left[2 + (t_1 + t_2)^2\right], \frac{a}{2}(t_1 + t_2)(1 - t_1 t_2)\right)$$

How to eliminate t_1, t_2 to obtain an equation in x and y only?

Note

$$\angle AOB = \frac{\pi}{2} \implies \frac{2at_1}{at_1^2} \cdot \frac{2at_2}{at_2^2} = -1 \implies t_1 t_2 = -4.$$

So

$$y = \frac{a}{2}(t_1 + t_2)(1 - (-4)) \implies t_1 + t_2 = \frac{2y}{5a}.$$

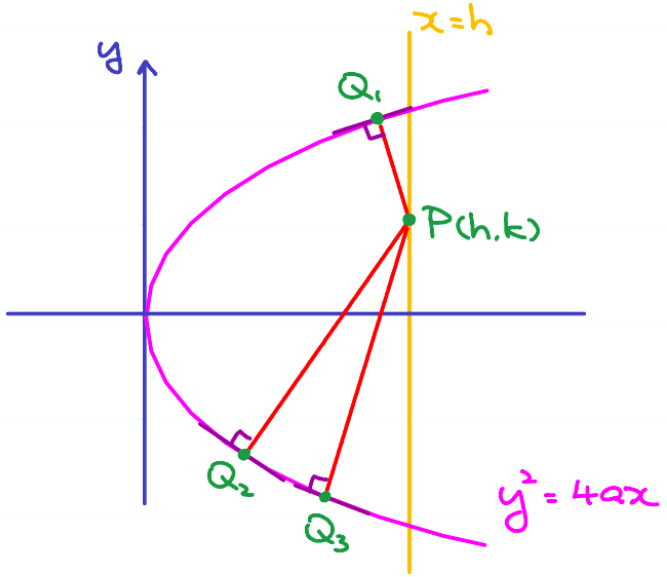
Put $t_1 + t_2 = \frac{2y}{5a}$ into $x = \frac{a}{2}\left[2 + (t_1 + t_2)^2\right]$, we have

$$y^2 = \frac{25a}{2}(x - a).$$

Exercise 11.2

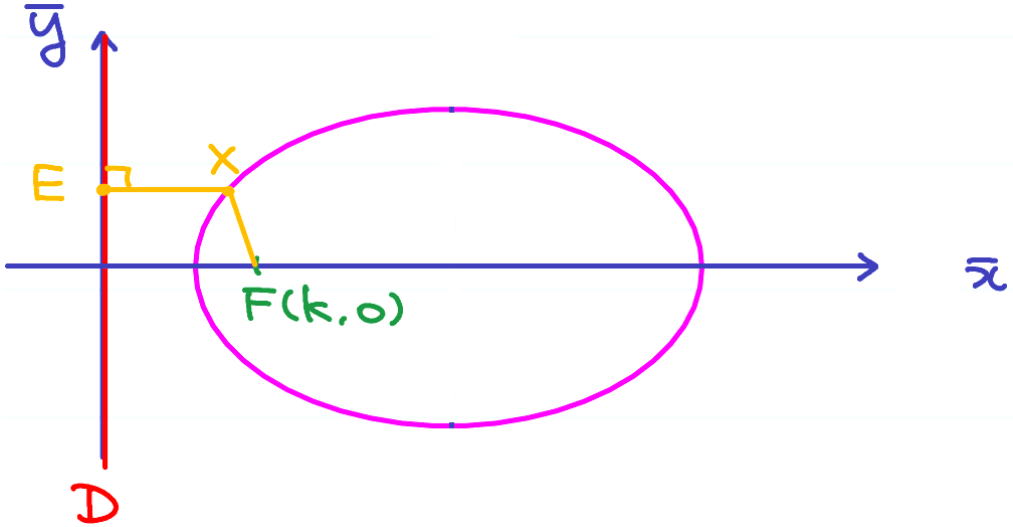
A point P moves on the line $x = h$ and the three normals from P meet the parabola $y^2 = 4ax$ at Q_1, Q_2, Q_3. Show that the centroid of $\triangle Q_1Q_2Q_3$ remains at a fixed point on the the x-axis.

(**Recall:** Centroid of (x_i, y_i), $i = 1, 2, 3$, is $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$.)



Ellipse $(0 < e < 1)$

Ellipse:
Assume $F = (k, 0)$, where $k > 0$, and $D : \bar{x} = 0$.



Let $X = (\bar{x}, \bar{y})$. Then

$$XF = e \cdot XE$$

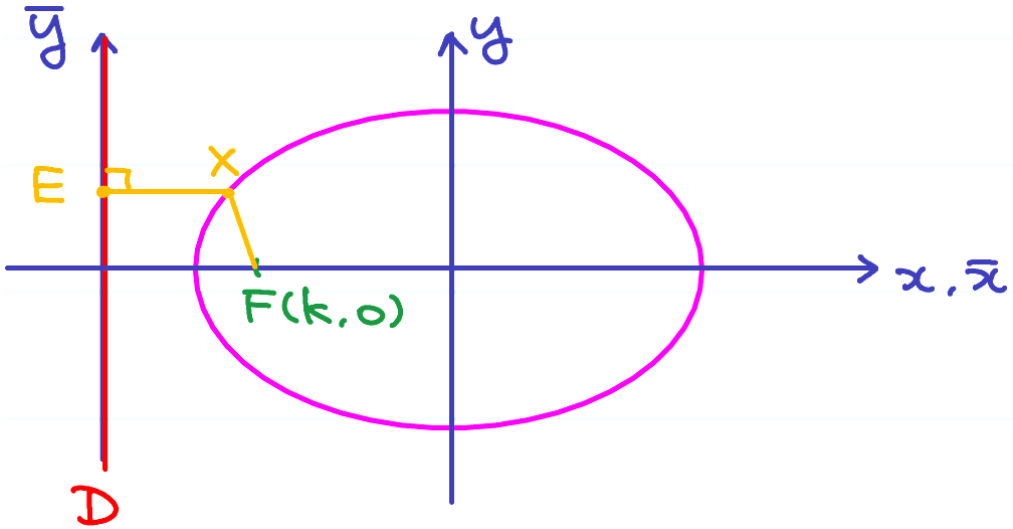
$$\sqrt{(\bar{x} - k)^2 + (\bar{y} - 0)^2} = e|\bar{x}|$$

$$\bar{x}^2 - 2k\bar{x} + \bar{y}^2 = e^2\bar{x}^2$$

$$(1 - e^2)\bar{x}^2 - 2k\bar{x} + \bar{y}^2 = 0$$

$$(1 - e^2)\left(\bar{x} - \frac{k}{1 - e^2}\right)^2 + \bar{y}^2 = \frac{k^2e^2}{1 - e^2}$$

Ellipse $(0 < e < 1)$



Let $x = \bar{x} - \frac{k}{1-e^2}$ and $y = \bar{y}$, the equation becomes

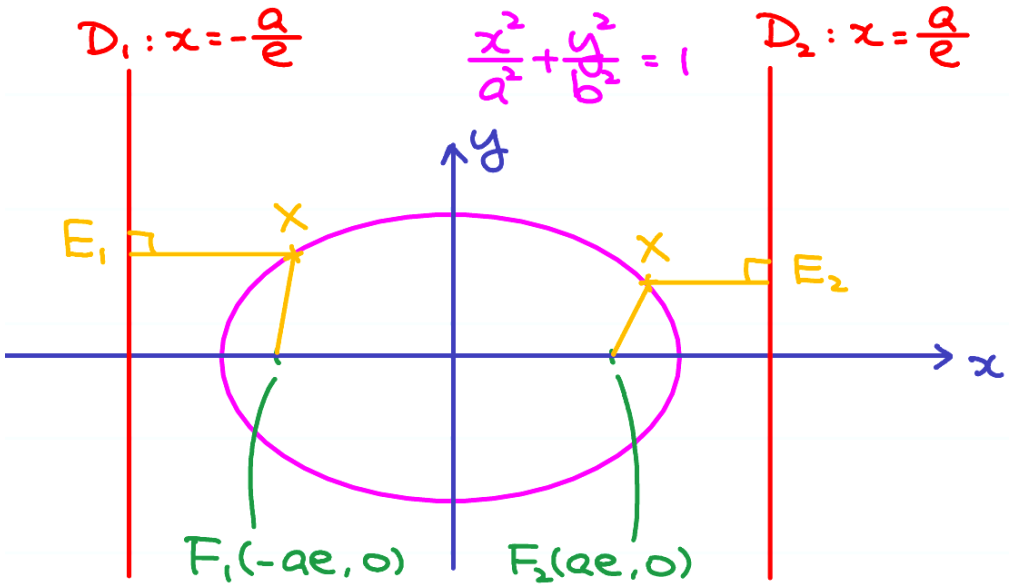
$$(1-e^2)x^2 + y^2 = \frac{k^2e^2}{1-e^2}$$

$$\frac{x^2}{(\frac{ke}{1-e^2})^2} + \frac{y^2}{(\frac{ke}{\sqrt{1-e^2}})^2} = 1.$$

Let $a = \frac{ke}{1-e^2}$, $b = a\sqrt{1-e^2}$ (and so $0 < b < a$),

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Ellipse $(0 < e < 1)$



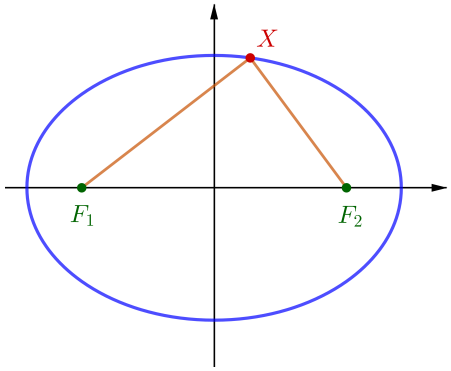
- a is called the **semimajor axis**, b the **semiminor axis** $(0 < b < a)$.
- Centre: $(0, 0)$, eccentricity: $e = \sqrt{1 - \frac{b^2}{a^2}}$.
- Directrix: $\bar{x} = 0 \quad \longleftrightarrow \quad x = 0 - \frac{k}{1-e^2} = -\frac{a}{e}$

  Focus: $(\bar{x}, \bar{y}) = (k, 0) \quad \longleftrightarrow \quad (x, y) = (-\frac{ke^2}{1-e^2}, 0) = (-ae, 0)$
- $XF_1 = e \cdot XE_1$ or $XF_2 = e \cdot XE_2$ will give the same ellipse.
- F_1 and F_2 are called the foci of the ellipse, D_1 and D_2 the directrices.

Ellipse ($0 < e < 1$)

Alternative Characterization of an ellipse:

An ellipse is the locus of a point X in a plane whose distances from two fixed points (the **foci**) in the plane have a constant sum.



Let $a > 0$ and $0 < e < 1$ be constants.
Suppose $F_1(-ae, 0)$ and $F_2(ae, 0)$ are the foci, and $X(x, y)$ is a point on plane such that $XF_1 + XF_2 = 2a$. Then $X(x, y)$ satisfies the equation

$$\sqrt{(x+ae)^2 + y^2} + \sqrt{(x-ae)^2 + y^2} = 2a \tag{1}$$

Multiplying by $\sqrt{(x+ae)^2 + y^2} - \sqrt{(x-ae)^2 + y^2}$, we have

$$\sqrt{(x+ae)^2 + y^2} - \sqrt{(x-ae)^2 + y^2} = 2ex. \tag{2}$$

$((1) + (2))^2$:

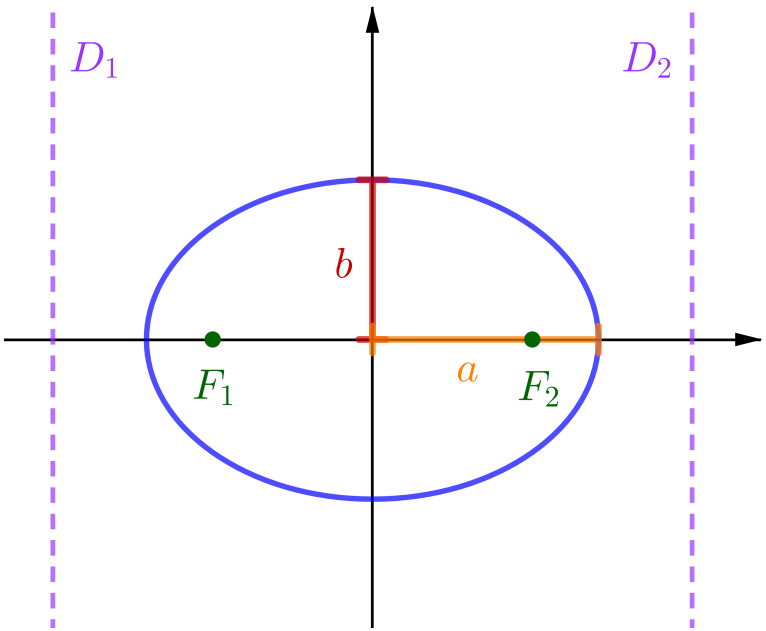
$$(x+ae)^2 + y^2 = (a+ex)^2$$
$$(1-e^2)x^2 + y^2 = a^2(1-e^2).$$

Let $b = a\sqrt{1-e^2}$, we have

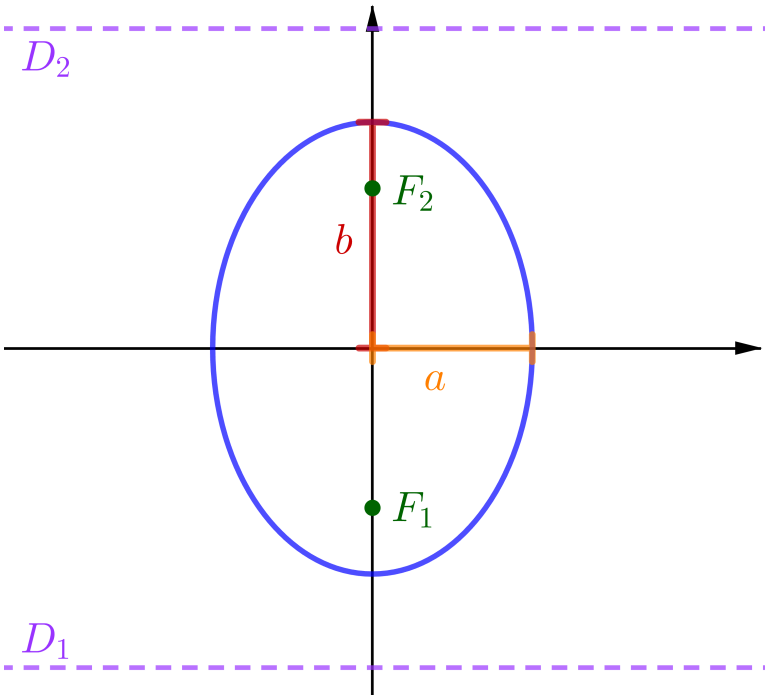
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Ellipse $(0 < e < 1)$

Two cases:



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,\ a > b > 0$$



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,\ b > a > 0$$

Ellipse $(0 < e < 1)$

Let $P(x_0, y_0)$ be a point on the ellipse $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
Differentiate both sides with respect to x:

$$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$$

$$\left.\frac{dy}{dx}\right|_{(x,y)=(x_0,y_0)} = -\frac{b^2 x_0}{a^2 y_0} \qquad (\text{if } y_0 \neq 0)$$

Tangent of $\mathcal{E}$ at $P(x_0, y_0)$:

$$\frac{y - y_0}{x - x_0} = -\frac{b^2 x_0}{a^2 y_0}$$

$$b^2 x_0 x + a^2 y_0 y - (b^2 x_0^2 + a^2 y_0^2) = 0$$

$$b^2 x_0 x + a^2 y_0 y - a^2 b^2 = 0 \quad (\because \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1)$$

Normal of $\mathcal{E}$ at $P(x_0, y_0)$:

$$\frac{y - y_0}{x - x_0} = \frac{a^2 y_0}{b^2 x_0}$$

$$a^2 y_0 x - b^2 x_0 y + (b^2 - a^2) x_0 y_0 = 0$$

Ellipse $(0 < e < 1)$

Parametrization:
The curve $\gamma : [0, 2\pi] \to \mathbb{R}^2$ defined by $\gamma(\theta) = (a\cos\theta, b\sin\theta)$ gives a parametrization of the ellipse $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
Let $P(a\cos\theta, b\sin\theta)$ be a point on the ellipse $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = -\frac{b\cos\theta}{a\sin\theta} \quad (\text{if } \theta \neq 0, \pi, 2\pi) \implies \left.\frac{dy}{dx}\right|_{\theta=\theta_0} = -\frac{b\cos\theta_0}{a\sin\theta_0}$$

Tangent of $\mathcal{E}$ at $P(a\cos\theta_0, b\sin\theta_0)$:

$$\frac{y - b\sin\theta_0}{x - a\cos\theta_0} = -\frac{b\cos\theta_0}{a\sin\theta_0}$$
$$(b\cos\theta_0)x + (a\sin\theta_0)y - ab(\cos^2\theta_0 + \sin^2\theta_0) = 0$$
$$(b\cos\theta_0)x + (a\sin\theta_0)y - ab = 0$$

Normal of $\mathcal{E}$ at $P(a\cos\theta_0, b\sin\theta_0)$:

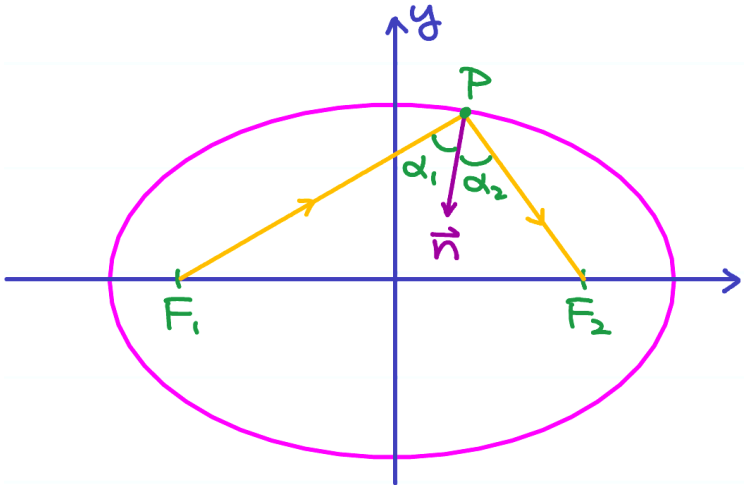
$$\frac{y - b\sin\theta_0}{x - a\cos\theta_0} = \frac{a\sin\theta_0}{b\cos\theta_0}$$
$$(a\sin\theta_0)x - (b\cos\theta_0)y + (b^2 - a^2)\sin\theta_0\cos\theta_0 = 0$$

Ellipse $(0 < e < 1)$

Example 11.5

Rays originated from one focus of the ellipse are reflected back to the other focus.

Let $P = (a\cos\theta, b\sin\theta)$ be a point on the ellipse $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
Then $\vec{n} = (-b\cos\theta)\vec{i} + (-a\sin\theta)\vec{j}$ is normal to $\mathcal{E}$ at P (pointing inwards).



$$\overrightarrow{PF_1} = (-ae - a\cos\theta)\vec{i} + (-b\sin\theta)\vec{j}$$

$$|\overrightarrow{PF_1}|^2 = a^2e^2 + 2a^2e\cos\theta + a^2\cos^2\theta + a^2(1-e^2)\sin^2\theta$$

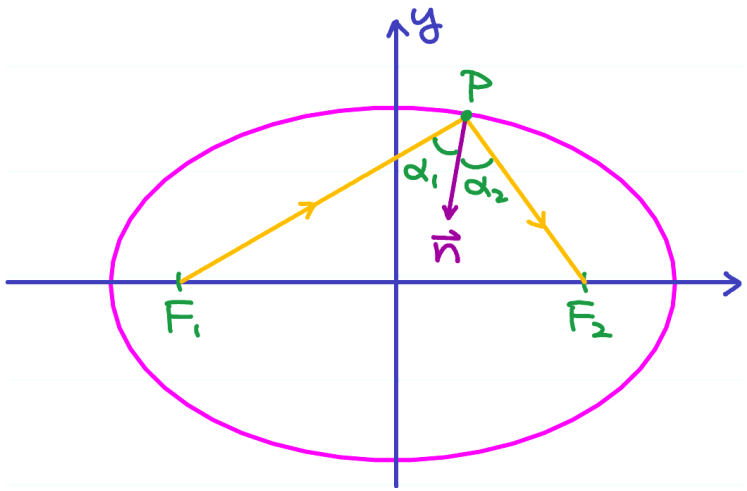
$$= a^2e^2\cos^2\theta + 2a^2e\cos\theta + a^2 = a^2(e\cos\theta + 1)^2$$

$$|\overrightarrow{PF_1}| = a(1 + e\cos\theta)$$

$$\cos\alpha_1 = \frac{\overrightarrow{PF_1}\cdot\vec{n}}{|\overrightarrow{PF_1}||\vec{n}|} = \frac{ab(1+e\cos\theta)}{a(1+e\cos\theta)|\vec{n}|} = \frac{b}{|\vec{n}|}$$

Ellipse $(0 < e < 1)$

Let $P = (a\cos\theta, b\sin\theta)$ be a point on the ellipse $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
Then $\vec{n} = (-b\cos\theta)\vec{i} + (-a\sin\theta)\vec{j}$ is normal to $\mathcal{E}$ at P (pointing inwards).



$$\overrightarrow{PF_2} = (ae - a\cos\theta)\vec{i} + (-b\sin\theta)\vec{j}$$

$$|\overrightarrow{PF_2}|^2 = a^2e^2 - 2a^2e\cos\theta + a^2\cos^2\theta + a^2(1-e^2)\sin^2\theta$$

$$= a^2e^2\cos^2\theta - 2a^2e\cos\theta + a^2 = a^2(e\cos\theta - 1)^2$$

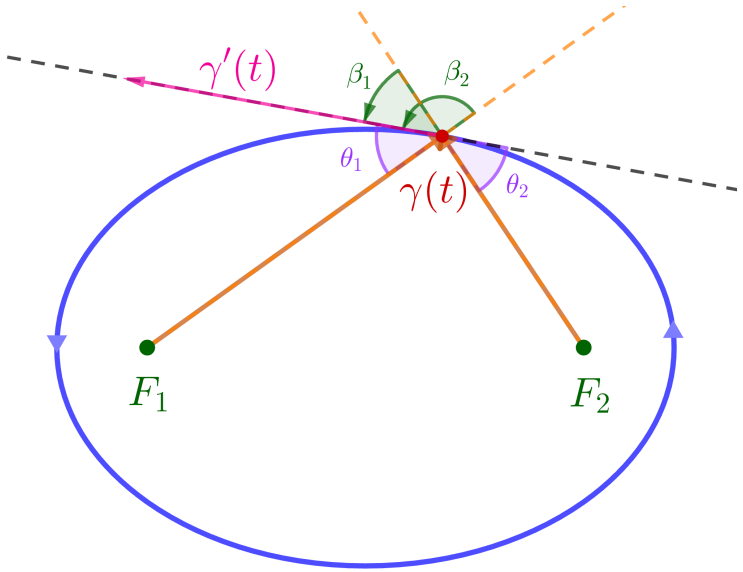
$$|\overrightarrow{PF_2}| = a(1 - e\cos\theta) \qquad (\because 0 < e < 1)$$

$$\cos\alpha_2 = \frac{\overrightarrow{PF_2}\cdot\vec{n}}{|\overrightarrow{PF_2}||\vec{n}|} = \frac{ab(1-e\cos\theta)}{a(1-e\cos\theta)|\vec{n}|} = \frac{b}{|\vec{n}|} = \cos\alpha_1$$

So $\alpha_1 = \alpha_2$.

Ellipse $(0 < e < 1)$

Alternative approach:



Let $\vec{a}, \vec{b}$ be the position vector of the two foci F_1, F_2 of the ellipse.
Then, the equation of the ellipse can be expressed in vector form as

$$|\vec{x} - \vec{a}| + |\vec{x} - \vec{b}| = c, \quad \text{a constant.}$$

Suppose $\vec{\gamma}(t)$ is a parametrization of the ellipse. We have

$$|\vec{\gamma}(t) - \vec{a}| + |\vec{\gamma}(t) - \vec{b}| = c.$$

Differentiate both sides with respect to t:

$$\vec{\gamma}'(t) \cdot \frac{\vec{\gamma}(t) - \vec{a}}{|\gamma(t) - \vec{a}|} + \vec{\gamma}'(t) \cdot \frac{\vec{\gamma}(t) - \vec{b}}{|\gamma(t) - \vec{b}|} = 0$$

$$\cos\beta_1 + \cos\beta_2 = 0$$

$$\beta_2 + \beta_1 = \pi$$

$$\therefore \theta_1 = \theta_2$$

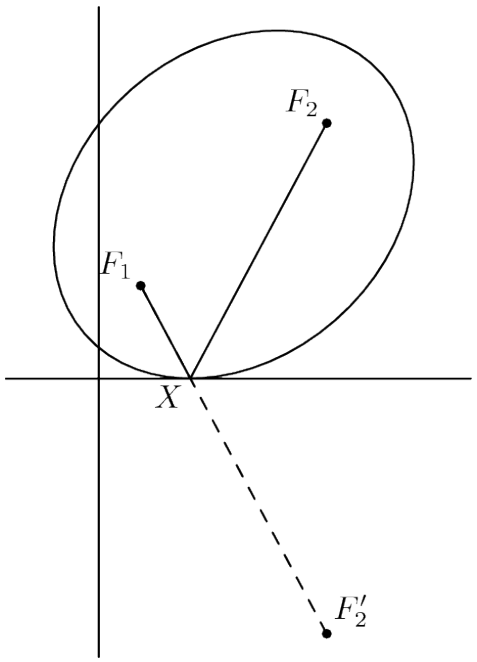
Example 11.6

An ellipse has foci at $(9, 20)$ and $(49, 55)$ in the xy-plane and is tangent to the x-axis. What is the length of its semi-major axis?

Let $F_1 = (9, 20)$, $F_2 = (49, 55)$ and let $X = (x, 0)$ be the point of tangency of the ellipse with the x-axis. Let $F_2' = (49, -55)$, the reflection of F_2 across the x-axis. By the reflection property of ellipses, F_1, X, F_2' are collinear. Thus

$$\begin{aligned} XF_1 + XF_2 &= XF_1 + XF_2' = F_1F_2' \\ &= \sqrt{(9-49)^2 + (20+55)^2} = 85. \end{aligned}$$

Hence, the length of its semi-major axis is $(XF_1 + XF_2)/2 = 85/2$.



Exercise 11.3

Let P, Q be points moving on an ellipse $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ such that $\angle POQ = \frac{\pi}{2}$ and the tangents at P, Q intersect at T. Find the equation of locus of T.

Let $P = (a\cos\theta_1, b\sin\theta_1), Q = (a\cos\theta_2, b\sin\theta_2)$.
Equations tangents at P, Q:

$$(b\cos\theta_i)x + (a\sin\theta_i)y - ab = 0, \quad i = 1, 2.$$

Solving the equations, we have

$$\begin{cases} x = \frac{a(\sin\theta_2 - \sin\theta_1)}{\sin\theta_2\cos\theta_1 - \sin\theta_1\cos\theta_2} = \frac{2a\sin\frac{\theta_2-\theta_1}{2}\cos\frac{\theta_2+\theta_1}{2}}{\sin(\theta_2-\theta_1)} \\ y = \frac{-b(\cos\theta_2 - \cos\theta_1)}{\sin\theta_2\cos\theta_1 - \sin\theta_1\cos\theta_2} = \frac{2b\sin\frac{\theta_2-\theta_1}{2}\sin\frac{\theta_2+\theta_1}{2}}{\sin(\theta_2-\theta_1)} \end{cases}.$$

Hence, the tangents at P, Q intersect at

$$T = \left(\frac{a\cos\frac{\theta_2+\theta_1}{2}}{\cos\frac{\theta_2-\theta_1}{2}}, \frac{b\sin\frac{\theta_2+\theta_1}{2}}{\cos\frac{\theta_2-\theta_1}{2}} \right).$$

Ellipse $(0 < e < 1)$

Since $\angle POQ = \frac{\pi}{2}$, we have

$$\begin{aligned}
\frac{b\sin\theta_1}{a\cos\theta_1}\cdot\frac{b\sin\theta_2}{a\cos\theta_2} &= -1\\
b^2\sin\theta_1\sin\theta_2 &= -a^2\cos\theta_1\cos\theta_2\\
-\frac{b^2}{2}[\cos(\theta_2-\theta_1)-\cos(\theta_2+\theta_1)] &= \frac{a^2}{2}[\cos(\theta_2-\theta_1)+\cos(\theta_2+\theta_1)]\\
(a^2+b^2)\cos(\theta_2-\theta_1) &= (b^2-a^2)\cos(\theta_2+\theta_1)\\
(a^2+b^2)[2\cos^2\frac{\theta_2-\theta_1}{2}-1] &= (b^2-a^2)[2\cos^2\frac{\theta_2+\theta_1}{2}-1]\\
(a^2+b^2)\cos^2\frac{\theta_2-\theta_1}{2} &= b^2\cos^2\frac{\theta_2+\theta_1}{2}+a^2[1-\cos^2\frac{\theta_2+\theta_1}{2}]\\
(\frac{1}{a^2}+\frac{1}{b^2})\cos^2\frac{\theta_2-\theta_1}{2} &= \frac{1}{a^4}(a^2\cos^2\frac{\theta_2+\theta_1}{2})+\frac{1}{b^4}(b^2\sin^2\frac{\theta_2+\theta_1}{2}).
\end{aligned}$$

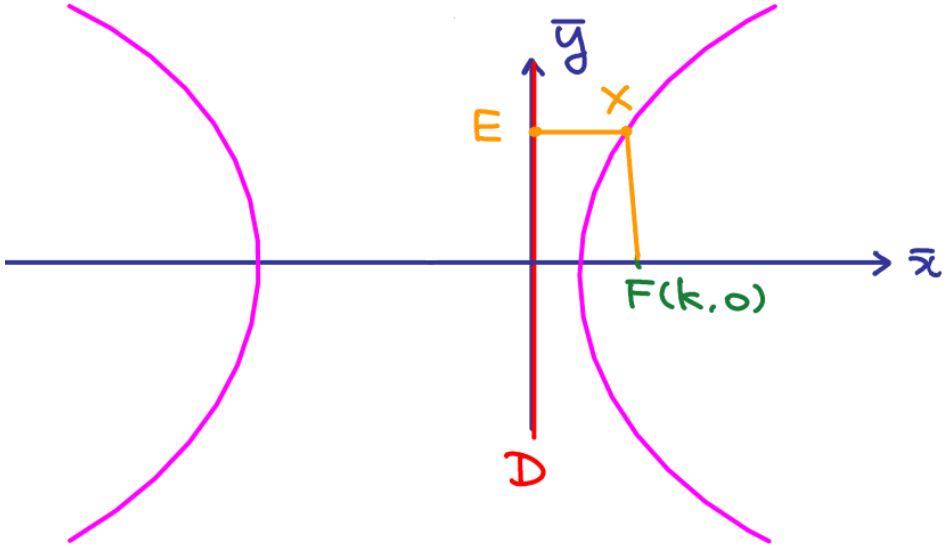
Hence, the coordinates of $T(x,y) = \left(\frac{a\cos\frac{\theta_2+\theta_1}{2}}{\cos\frac{\theta_2-\theta_1}{2}}, \frac{b\sin\frac{\theta_2+\theta_1}{2}}{\cos\frac{\theta_2-\theta_1}{2}}\right)$ satisfies

$$\frac{x^2}{a^4}+\frac{y^2}{b^4} = \frac{1}{a^2}+\frac{1}{b^2}.$$

GeoGebra Illustration

Hyperbola ($e > 1$)

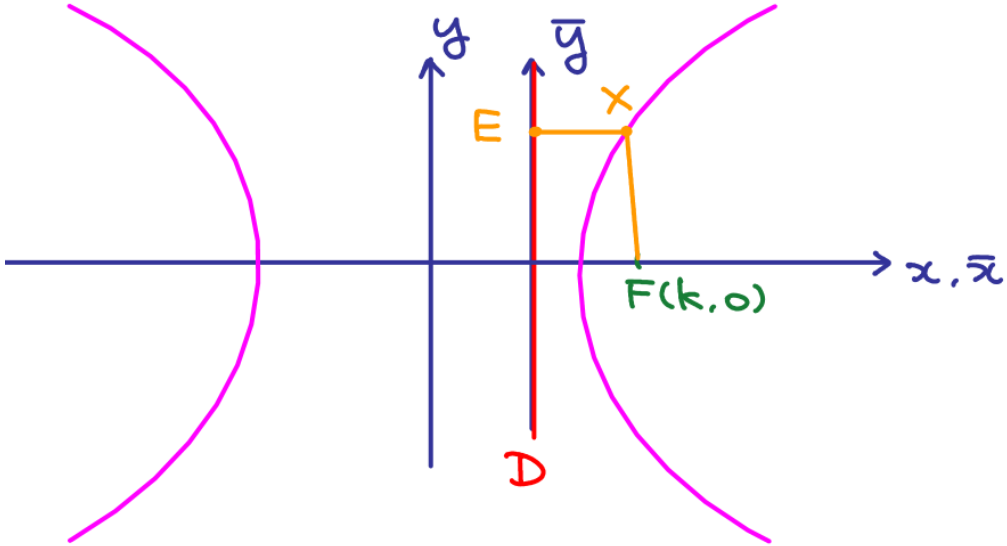
Hyperbola:
Assume $F = (k, 0)$, where $k > 0$, and $D : \bar{x} = 0$.



Let $X = (\bar{x}, \bar{y})$. Then

$$
\begin{aligned}
XF &= e \cdot XE \\
\sqrt{(\bar{x} - k)^2 + (\bar{y} - 0)^2} &= e|\bar{x}| \\
\bar{x}^2 - 2k\bar{x} + k^2 + \bar{y}^2 &= e^2\bar{x}^2 \\
\bar{y}^2 &= (e^2 - 1)\bar{x}^2 + 2k\bar{x} - k^2 \\
\bar{y}^2 &= (e^2 - 1)\left(\bar{x} + \frac{k}{e^2 - 1}\right)^2 - \frac{k^2e^2}{e^2 - 1}
\end{aligned}
$$

Hyperbola ($e > 1$)



Let $x = \bar{x} + \frac{k}{e^2-1}$ and $y = \bar{y}$, the eq. becomes

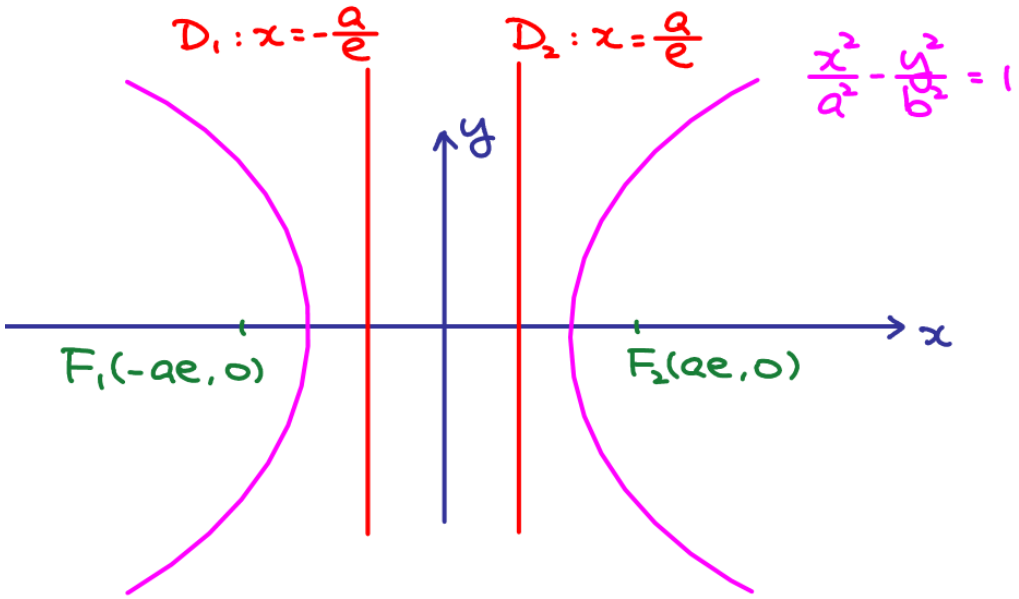
$$y^2 = (e^2-1)x^2 - \frac{k^2e^2}{e^2-1}$$

$$\frac{x^2}{(\frac{ke}{e^2-1})^2} - \frac{y^2}{(\frac{ke}{\sqrt{e^2-1}})^2} = 1.$$

Let $a = \frac{ke}{e^2-1}$, $b = a\sqrt{e^2-1}$,

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

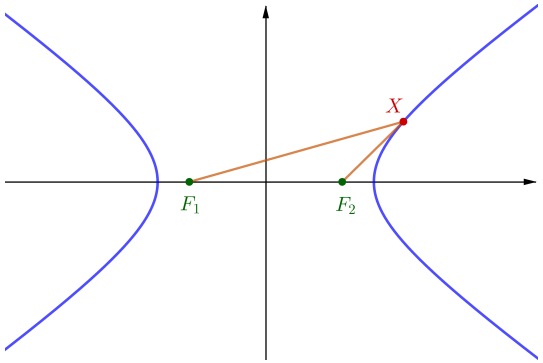
Hyperbola ($e > 1$)



- Centre: $(0,0)$, eccentricity: $e = \sqrt{1 + \frac{b^2}{a^2}}$.
- Directrix: $\bar{x} = 0 \quad \longleftrightarrow \quad x = 0 + \frac{k}{e^2-1} = \frac{a}{e}$

  Focus: $(\bar{x}, \bar{y}) = (k, 0) \quad \longleftrightarrow \quad (x, y) = (\frac{ke^2}{e^2-1}, 0) = (ae, 0)$
- $XF_1 = e \cdot XE_1$ or $XF_2 = e \cdot XE_2$ will give the same hyperbola.
- F_1 and F_2 are called the foci of the hyperbola, D_1 and D_2 the directrices.

Alternative Characterization of a hyperbola:



A hyperbola is the locus of a point X in a plane whose distances from two fixed points (the **foci**) in the plane have a constant difference.

Let $a > 0$ and $e > 1$ be constants.
Suppose $F_1(-ae, 0)$ and $F_2(ae, 0)$ are the foci, and $|XF_1 - XF_2| = 2a$, where $X(x, y)$ is a point on the hyperbola. Then $X(x, y)$ satisfies the equation

$$\sqrt{(x+ae)^2 + y^2} - \sqrt{(x-ae)^2 + y^2} = \pm 2a. \tag{1}$$

Multiplying by $\sqrt{(x+ae)^2 + y^2} + \sqrt{(x-ae)^2 + y^2}$, we have

$$\sqrt{(x+ae)^2 + y^2} + \sqrt{(x-ae)^2 + y^2} = \pm 2ex. \tag{2}$$

$((1) + (2))^2$:

$$(x+ae)^2 + y^2 = [\pm(a+ex)]^2$$
$$(e^2-1)x^2 - y^2 = a^2(e^2-1).$$

Let $b = a\sqrt{1-e^2}$, we have

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Hyperbola ($e > 1$)

Let $P(x_0, y_0)$ be a point on the hyperbola $\mathcal{H} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.
Differentiate both sides with respect to x:

$$\frac{2x}{a^2} - \frac{2y}{b^2}\frac{dy}{dx} = 0$$

$$\left.\frac{dy}{dx}\right|_{(x,y)=(x_0,y_0)} = \frac{b^2 x_0}{a^2 y_0} \qquad (\text{if } y_0 \neq 0)$$

Tangent of $\mathcal{H}$ at $P(x_0, y_0)$:

$$\frac{y - y_0}{x - x_0} = \frac{b^2 x_0}{a^2 y_0}$$

$$b^2 x_0 x - a^2 y_0 y - (b^2 x_0^2 - a^2 y_0^2) = 0$$

$$b^2 x_0 x - a^2 y_0 y - a^2 b^2 = 0 \quad (\because \frac{x_0^2}{a^2} - \frac{y_0^2}{b^2} = 1)$$

Normal of $\mathcal{H}$ at $P(x_0, y_0)$:

$$\frac{y - y_0}{x - x_0} = -\frac{a^2 y_0}{b^2 x_0}$$

$$a^2 y_0 x + b^2 x_0 y - (a^2 + b^2) x_0 y_0 = 0$$

Hyperbola ($e > 1$)

Parametrization:
The curve $\gamma : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}^2$ (resp. $\gamma : (\frac{\pi}{2}, \frac{3\pi}{2}) \to \mathbb{R}^2$) defined by $\gamma(\theta) = (a\sec\theta, b\tan\theta)$ gives a parametrization of the right (resp. left) half of the hyperbola $\mathcal{H} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.
Let $P(a\sec\theta, b\tan\theta)$ be a point on the hyperbola $\mathcal{H} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{b\sec^2\theta}{a\sec\theta\tan\theta} = \frac{b\sec\theta}{a\tan\theta} \quad (\text{if } \theta \neq 0, \pi) \implies \left.\frac{dy}{dx}\right|_{\theta=\theta_0} = \frac{b\sec\theta_0}{a\tan\theta_0}$$

Tangent of $\mathcal{H}$ at $P(a\sec\theta_0, b\tan\theta_0)$:

$$\frac{y - b\tan\theta_0}{x - a\sec\theta_0} = \frac{b\sec\theta_0}{a\tan\theta_0}$$
$$(b\sec\theta_0)x - (a\tan\theta_0)y - ab = 0$$

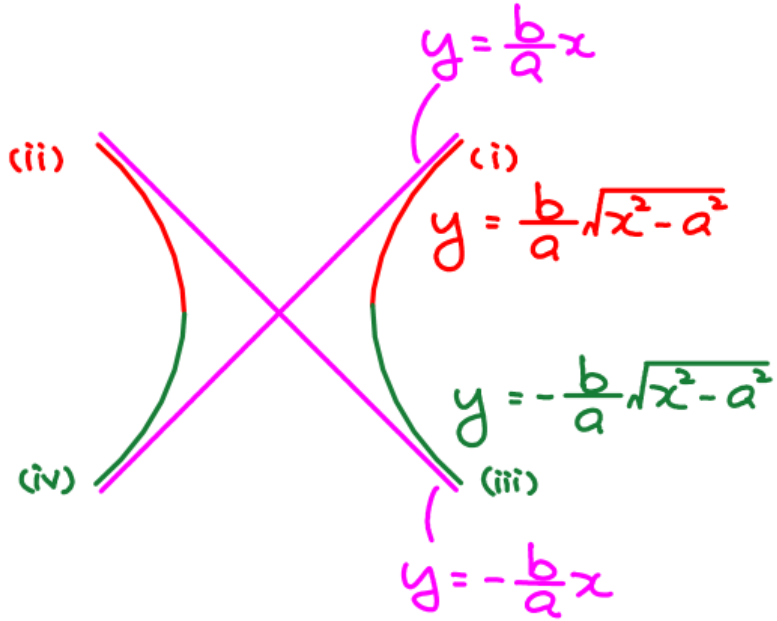
Normal of $\mathcal{H}$ at $P(a\sec\theta_0, b\tan\theta_0)$:

$$\frac{y - b\tan\theta_0}{x - a\sec\theta_0} = -\frac{a\tan\theta_0}{b\sec\theta_0}$$
$$(a\tan\theta_0)x + (b\sec\theta_0)y - (a^2 + b^2)\sec\theta_0\tan\theta_0 = 0$$

Hyperbola ($e > 1$)

Asymptote of $\mathcal{H}$: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm\frac{b}{a}x$

Write $y^2 = \frac{b^2}{a^2}(x^2 - a^2) \implies y = \pm\frac{b}{a}\sqrt{x^2 - a^2}$



Case 1: $y = \frac{b}{a}\sqrt{x^2 - a^2}$

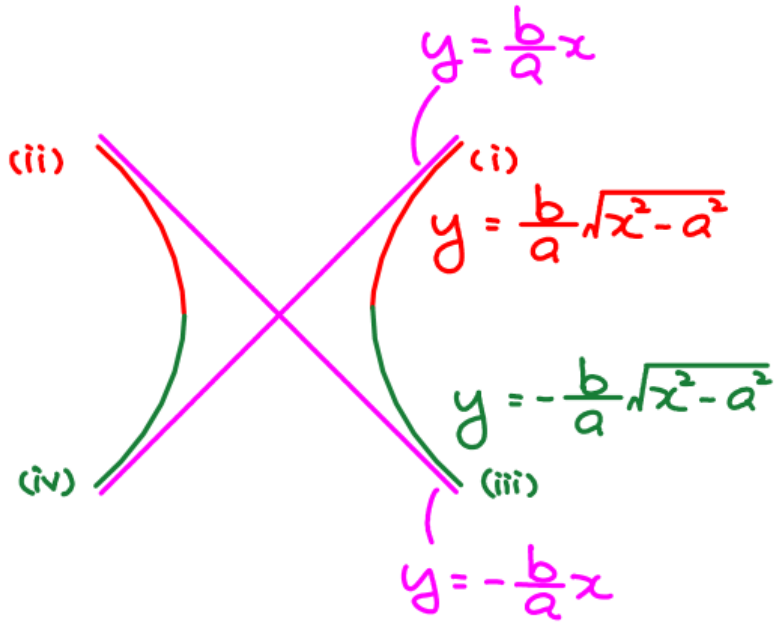
(i) $$\lim_{x\to+\infty} y - \frac{b}{a}x = \lim_{x\to+\infty} \frac{b}{a}(\sqrt{x^2 - a^2} - \sqrt{x^2}) = \lim_{x\to+\infty} \frac{b}{a}\frac{(-a^2)}{\sqrt{x^2 - a^2} + \sqrt{x^2}} = 0$$

(ii) $$\lim_{x\to-\infty} y + \frac{b}{a}x = \lim_{x\to-\infty} \frac{b}{a}(\sqrt{x^2 - a^2} - \sqrt{x^2}) = \lim_{x\to-\infty} \frac{b}{a}\frac{(-a^2)}{\sqrt{x^2 - a^2} + \sqrt{x^2}} = 0$$

Caution: $x = -\sqrt{x^2}$ when $x < 0$.

Hyperbola ($e > 1$)

Asymptote of $\mathcal{H}$: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $y = \pm\frac{b}{a}x$



Case 2: $y = -\frac{b}{a}\sqrt{x^2 - a^2}$

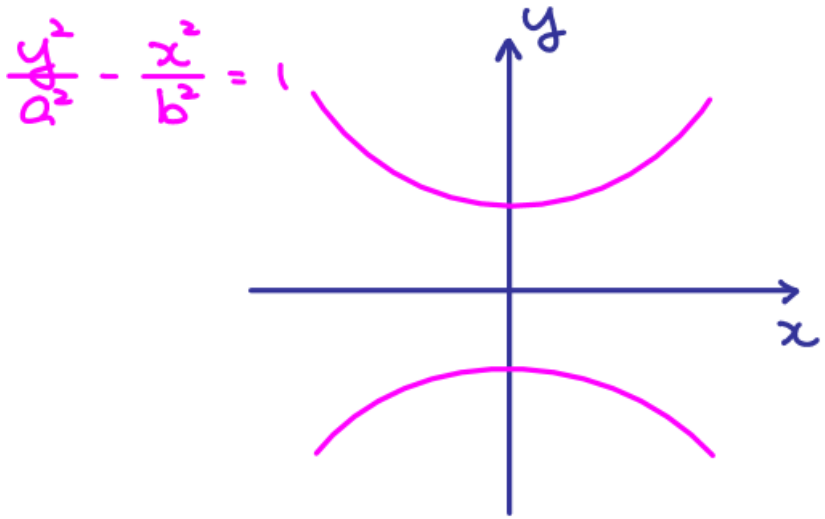
(iii) $$\lim_{x\to+\infty} y + \frac{b}{a}x = \lim_{x\to+\infty} \frac{b}{a}(\sqrt{x^2} - \sqrt{x^2 - a^2}) = \lim_{x\to+\infty} \frac{b}{a}\frac{a^2}{\sqrt{x^2} + \sqrt{x^2 - a^2}} = 0$$

(iv) $$\lim_{x\to-\infty} y - \frac{b}{a}x = \lim_{x\to-\infty} \frac{b}{a}(+\sqrt{x^2} - \sqrt{x^2 - a^2}) = \lim_{x\to-\infty} \frac{b}{a}\frac{a^2}{\sqrt{x^2} + \sqrt{x^2 - a^2}} = 0$$

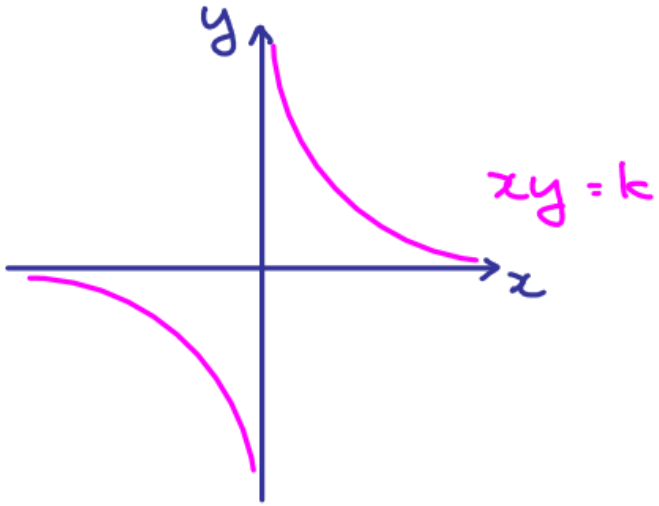
Caution: $x = -\sqrt{x^2}$ when $x < 0$.

Hyperbola ($e > 1$)

Other forms:
Interchange the roles of x and y:



Take $a = b$ and rotate by $\pi/4$:

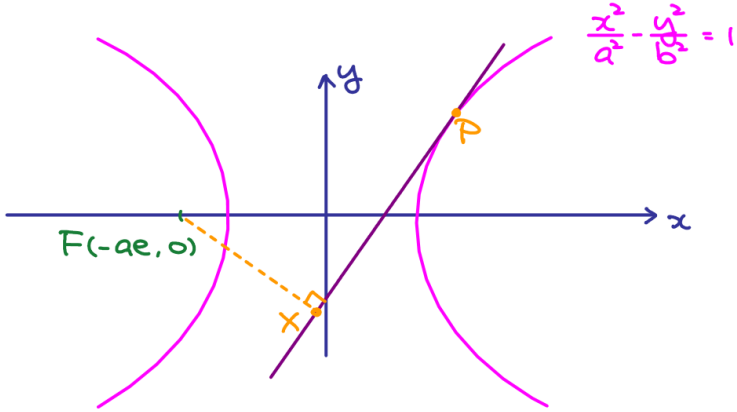


Example 11.7

Show that the locus of the foot of perpendicular X from the focus $F(-ae, 0)$ to the tangent of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a circle.

Let $P(a\sec\theta, b\tan\theta)$ be a movable point on the hyperbola.
Equation of tangent at P:

$$(b\sec\theta)x - (a\tan\theta)y = ab \qquad (1)$$



$$\text{Slope of } XF = -\frac{a\tan\theta}{b\sec\theta} = \frac{y-0}{x-(-ae)}$$

$$(a\tan\theta)x + (b\sec\theta)y = -a^2 e\tan\theta \qquad (2)$$

$(1)^2 + (2)^2$:

$$[(b\sec\theta)x - (a\tan\theta)y]^2 + [(a\tan\theta)x + (b\sec\theta)y]^2 = a^2b^2 + a^4e^2\tan^2\theta$$

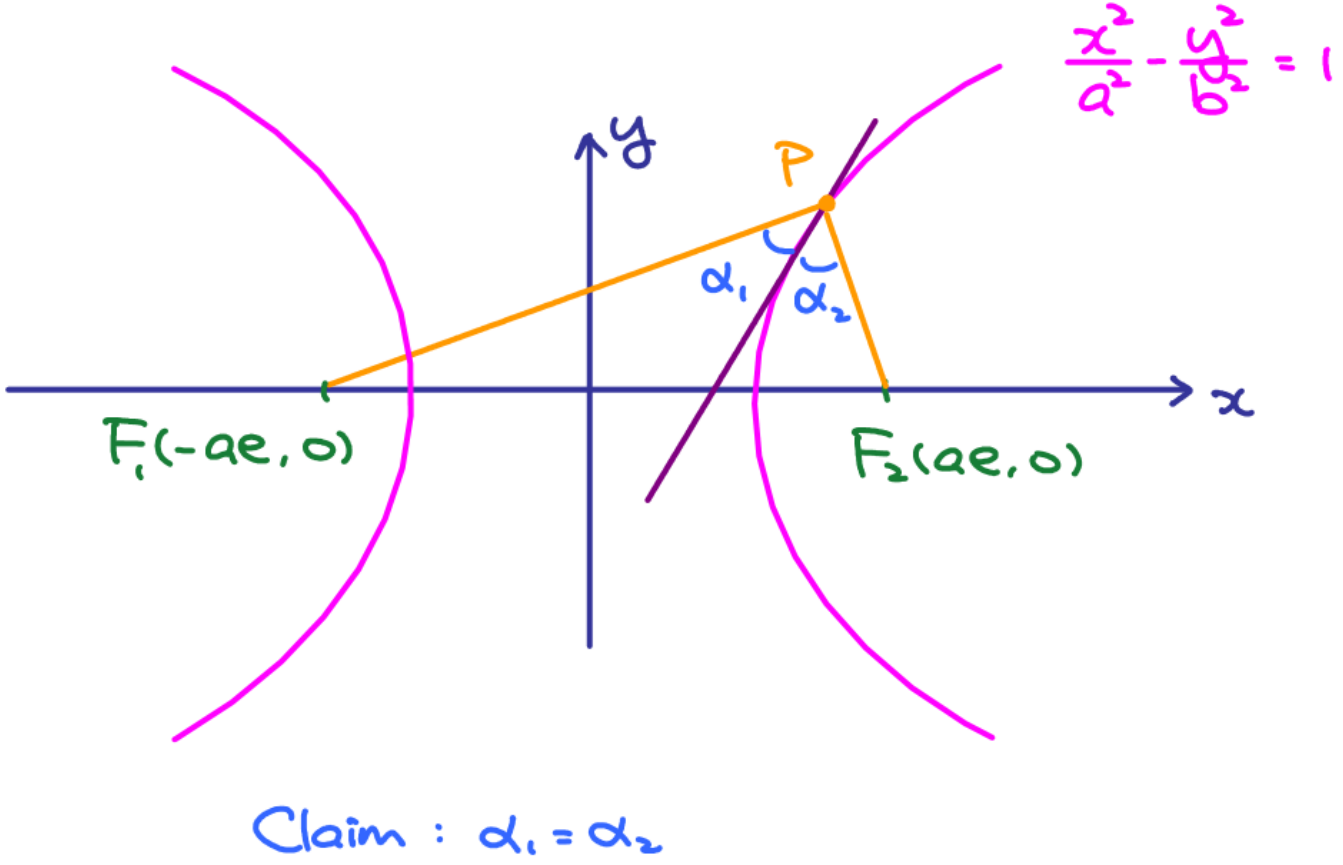
$$(x^2+y^2)(a^2\tan^2\theta + b^2\sec^2\theta) = a^2b^2 + a^2(a^2+b^2)\tan^2\theta$$

$$= a^2(a^2\tan^2\theta + b^2\sec^2\theta)$$

So $x^2 + y^2 = a^2$.

Example 11.8

Let P be a point on a hyperbola. Suppose F_1, F_2 are foci of the hyperbola. Show that PF_1 and PF_2 are equally inclined to the tangent at P.



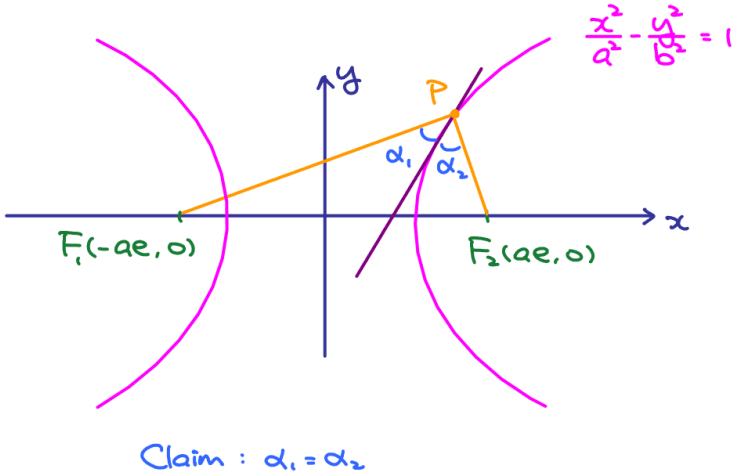
Hyperbola ($e > 1$)

Example 11.8

Let P be a point on a hyperbola. Suppose F_1, F_2 are foci of the hyperbola. Show that PF_1 and PF_2 are equally inclined to the tangent at P.

Let $\vec{a}, \vec{b}$ be the position vector of the two foci F_1, F_2 of the hyperbola.
Then, the equation of one half of the hyperbola can be expressed in vector form as

$$|\vec{x} - \vec{a}| - |\vec{x} - \vec{b}| = c, \quad \text{a constant.}$$



Suppose $\vec{\gamma}(t)$ is a parametrization of (one half of) the hyperbola. We have

$$|\vec{\gamma}(t) - \vec{a}| - |\vec{\gamma}(t) - \vec{b}| = c.$$

Differentiate both sides with respect to t:

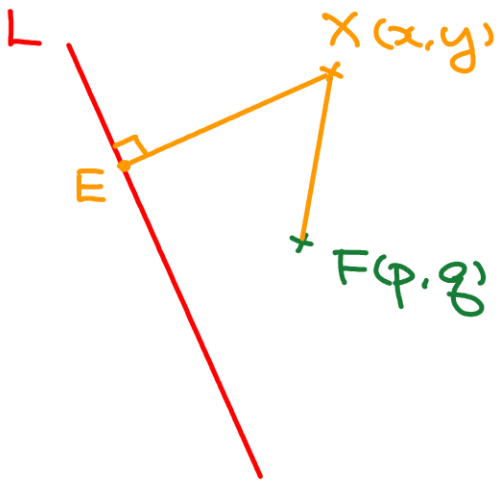
$$\vec{\gamma}'(t) \cdot \frac{\vec{\gamma}(t) - \vec{a}}{|\vec{\gamma}(t) - \vec{a}|} - \vec{\gamma}'(t) \cdot \frac{\vec{\gamma}(t) - \vec{b}}{|\vec{\gamma}(t) - \vec{b}|} = 0$$

$$\cos\alpha_1 - \cos\alpha_2 = 0$$

$$\therefore \alpha_1 = \alpha_2$$

General Conics and Their Classification (Optional)

Let $L : ax + by + c = 0$ be a directrix and $F(p, q)$ be a focus. Let $X(x, y)$ be a point.



The Focus-Directrix equation becomes

$$XF = e \cdot XE$$

$$\sqrt{(x-p)^2 + (y-q)^2} = e \cdot \left| \frac{ax + by + c}{\sqrt{a^2 + b^2}} \right|$$

$$(x-p)^2 + (y-q)^2 = e^2 \cdot \left| \frac{ax + by + c}{\sqrt{a^2 + b^2}} \right|^2 .$$

So a conic section must satisfy an equation of the form:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

General Conics and Their Classification (Optional)

Question

Does $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ always give a conic section?

Idea: Rotate the coordinate axes and use $x'y'$-coordinates to get rid of the cross term.
The equation then becomes

$$\underbrace{A'x'^2 + C'y'^2}_{\text{quadratic}} + \underbrace{D'x' + E'y' + F'}_{\text{linear}} = 0.$$

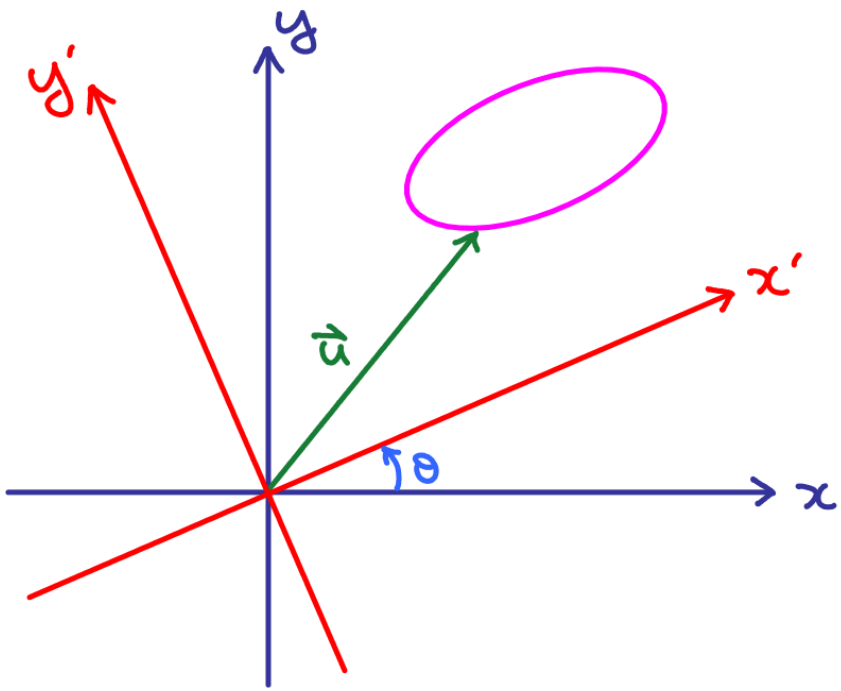
By completing the squares, the equation can be further reduced to

$$A'(x' - h)^2 + C'(y' - k)^2 = m.$$

The signs of A, C and m will then determine whether the equation is an ellipse, a parabola, a hyperbola, or a degenerate conic.

General Conics and Their Classification (Optional)

Rotate the x, y coordinate axes by θ in the couterclockwise direction to obtain the new x', y' coordinate axes.



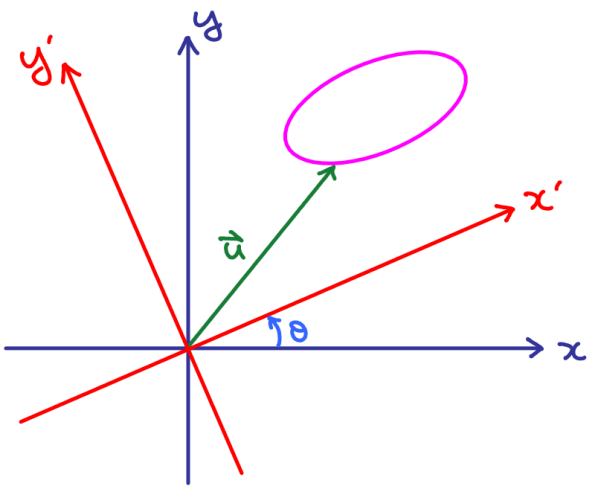
Let $\vec{e}_1$, $\vec{e}_2$ be the unit coordinate vectors in $x'y'$-coordinates. Then

$$\vec{e}_1 = \cos\theta\vec{i} + \sin\theta\vec{j}, \quad \vec{e}_2 = -\sin\theta\vec{i} + \cos\theta\vec{j}.$$

General Conics and Their Classification (Optional)

A point $\vec{u}$ can be expressed as:

$$\vec{u} = x\vec{i} + y\vec{j} \quad \text{and}$$
$$\vec{u} = x'\vec{e}_1 + y'\vec{e}_2$$
$$= x'(\cos\theta\vec{i} + \sin\theta\vec{j}) + y'(-\sin\theta\vec{i} + \cos\theta\vec{j})$$
$$= (x'\cos\theta - y'\sin\theta)\vec{i} + (x'\sin\theta + y'\cos\theta)\vec{j}.$$



$$\therefore \begin{cases} x = x'\cos\theta - y'\sin\theta \\ y = x'\sin\theta + y'\cos\theta \end{cases} \qquad \text{or} \qquad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

Substitute the above into $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$:

$$A(x'\cos\theta - y'\sin\theta)^2 + B(x'\cos\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta)$$
$$+C(x'\sin\theta + y'\cos\theta)^2 + D(x'\cos\theta - y'\sin\theta) + E(x'\sin\theta + y'\cos\theta) + F = 0$$
$$(A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta)x'^2 + [-2A\sin\theta\cos\theta + B(\cos^2\theta - \sin^2\theta) + 2C\sin\theta\cos\theta]x'y'$$
$$+(A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta)y'^2 + (D\cos\theta + E\sin\theta)x' + (-D\sin\theta + E\cos\theta)y' + F = 0$$

Denote the last equation as

$$A'x'^2 + B'x'y' + C'y'^2 + D'x' + E'y' + F' = 0.$$

General Conics and Their Classification (Optional)

In particular,

$$\begin{cases} A' = A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta \\ B' = -2A\sin\theta\cos\theta + B(\cos^2\theta - \sin^2\theta) + 2C\sin\theta\cos\theta \\ C' = A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta. \end{cases}$$

- **Claim 1:** $A + C = A' + C'$

$$\begin{aligned} A' + C' &= (A\cos^2\theta + B\sin\theta\cos\theta + C\sin^2\theta) + (A\sin^2\theta - B\sin\theta\cos\theta + C\cos^2\theta) \\ &= (A + C)(\cos^2\theta + \sin^2\theta) = A + C \end{aligned}$$

- **Claim 2:** $B^2 - 4AC = B'^2 - 4A'C'$

$$\begin{aligned} B'^2 - 4A'C' &= B'^2 - (A' + C')^2 + (A' - C')^2 \\ &= [B\cos 2\theta - (A - C)\sin 2\theta]^2 - (A + C)^2 + [(A - C)\cos 2\theta + B\sin 2\theta]^2 \\ &= B^2 - (A + C)^2 + (A - C)^2 \\ &= B^2 - 4AC \end{aligned}$$

General Conics and Their Classification (Optional)

Solve θ such that $B' = 0$:

$$B' := -2A\sin\theta\cos\theta + B(\cos^2\theta - \sin^2\theta) + 2C\sin\theta\cos\theta = 0.$$

That is,

$$B\cos 2\theta - (A - C)\sin 2\theta = 0$$

- If $A \neq C$, take $\tan 2\theta = \dfrac{B}{A - C}$ which as a solution in $(-\dfrac{\pi}{4}, \dfrac{\pi}{4})$.
- If $A = C$, take $\theta = \dfrac{\pi}{4}$.

Therefore, it suffices to look at equation in form of

$$Ax^2 + Cy^2 + Dx + Ey + F = 0 \qquad (*)$$

- **Case 1** $(A = C = 0)$**:**
  $(*)$ becomes a linear equation $\quad Dx + Ey + F = 0$, which gives a straight line.

General Conics and Their Classification (Optional)

- **Case 2** $(A \neq 0, C = E = 0)$**:**
  $(*)$ becomes $\quad Ax^2 + Dx + F = 0$. Let $\Delta = D^2 - 4AF$.
  - $\Delta > 0$: It gives a pair of parallel lines $x = r_1$, $x = r_2$.
  - $\Delta = 0$: It gives a line $x = r$.
  - $\Delta < 0$: No solution.

- **Case 3** $(C \neq 0, A = D = 0)$**:**
  $(*)$ becomes $\quad Cy^2 + Ey + F = 0$. Similar to case 2.

- **Case 4** $(A \neq 0, C = 0, E \neq 0)$**:**
  $(*)$ becomes $\quad Ax^2 + Dx + Ey + F = 0$
  $$A(x-h)^2 = -E(y-k), \qquad \text{where } D = -2Ah, F = Ah^2 - Ek$$
  which is a parabola.

- **Case 5** $(C \neq 0, A = 0, D \neq 0)$**:**
  $(*)$ becomes $\quad Cy^2 + Dx + Ey + F = 0$. Similar to case 4.

- **Case 6** $(A \neq 0, C \neq 0)$**:**
  By completing the square, $(*)$ becomes $A(x-h)^2 + C(y-k)^2 = m$, where
  $$D = -2Ah, \quad E = -2Ck, \quad F = Ah^2 + Ck^2 - m.$$

  - If $AC > 0$, then A, C have the same sign.
    WLOG, assume $A, C > 0$. Then, $(\frac{x-h}{\sqrt{C}})^2 + (\frac{y-k}{\sqrt{A}})^2 = \frac{m}{AC}$.
    - $m > 0$: If $A = C$, it defines a circle; otherwise, an ellipse.
    - $m = 0$: It gives a single point (h, k).
    - $m < 0$: No solution.
  - If $AC < 0$, then A, C have opposite signs.
    WLOG, assume $A > 0, C < 0$. Then $(\frac{x-h}{\sqrt{-C}})^2 - (\frac{y-k}{\sqrt{A}})^2 = -\frac{m}{AC}$.
    - $m \neq 0$: It gives a hyperbola.
    - $m = 0$: It gives a pair of intersecting lines: $\frac{x-h}{\sqrt{-C}} = \pm\frac{y-k}{\sqrt{A}}$.

General Conics and Their Classification (Optional)

Summary:
Curve defined by $\quad Ax^2 + Cy^2 + Dx + Ey + F = 0$:

	$-4AC < 0$		$-4AC = 0$	$-4AC > 0$
	$A = C$	$A \neq C$		
Non-degenerate	circle	ellipse	parabola	hyperbola
Degenerate	single point empty set	single point empty set	2 parallel lines one double line empty set	2 intersecting lines

In general, by changing coordinates, we write

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

into the form

$$A'x'^2 + B'x'y' + C'y'^2 + D'x' + E'y' + F' = 0 \quad \text{with } B' = 0.$$

Then, $B^2 - 4AC = B'^2 - 4A'C' = -4A'C'$.
Furthermore, $A' = C' \implies B = 0 \quad (\because B^2 + (A - C)^2 = B'^2 + (A' - C')^2)$.

General Conics and Their Classification (Optional)

Proposition 11.1

$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ *represents*

(1) *a circle if* $B = 0$ *and* $A = C \neq 0$,

(2) *an ellipse if* $B^2 - 4AC < 0$,

(3) *a parabola if* $B^2 - 4AC = 0$,

(4) *a hyperbola if* $B^2 - 4AC > 0$.

Example 11.9

Determine the type of conic section defined by

$$5x^2 - 4xy + 8y^2 + 4\sqrt{5}x - 16\sqrt{5} + 4 = 0.$$

Sketch the curve.

Since $B^2 - 4AC = (-4)^2 - 4(5)(8) = -144 < 0$, so the conic is an ellipse.
Set $\tan 2\theta = \dfrac{B}{A-C} = \dfrac{-4}{5-8} = \dfrac{4}{3}$, where $-\dfrac{\pi}{4} < \theta < \dfrac{\pi}{4}$.

$$\frac{2\tan\theta}{1-\tan^2\theta} = \frac{4}{3}$$

$$2\tan^2\theta + 3\tan\theta - 2 = 0$$

$$\tan\theta = \frac{1}{2} \quad \text{or} \quad \tan\theta = -2 \text{ (rejected)}$$

$$\therefore \cos\theta = \frac{2}{\sqrt{5}}, \quad \sin\theta = \frac{1}{\sqrt{5}}$$

Example 11.9

Determine the type of conic section defined by

$$5x^2 - 4xy + 8y^2 + 4\sqrt{5}x - 16\sqrt{5} + 4 = 0.$$

Sketch the curve.

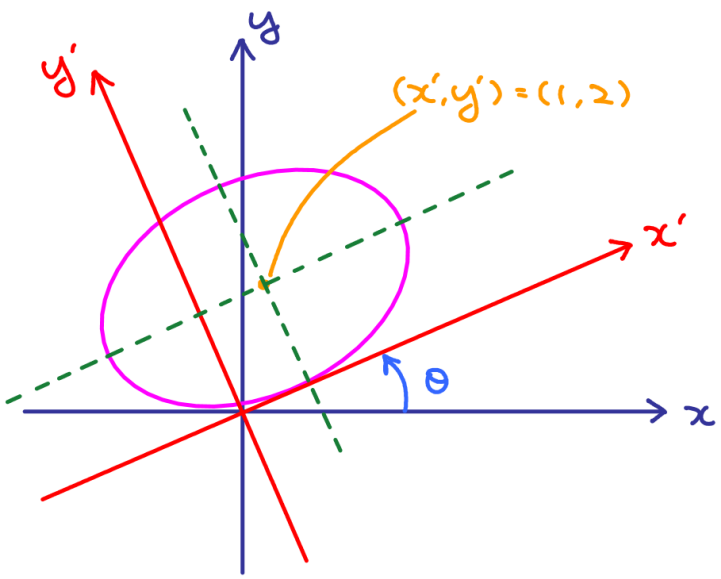
So $\begin{cases} x = x'\cos\theta - y'\sin\theta = \dfrac{1}{\sqrt{5}}(2x' - y') \\ y = x'\sin\theta + y'\cos\theta = \dfrac{1}{\sqrt{5}}(x' + 2y'). \end{cases}$

Put them into the given equation:

$$4x'^2 + 9y'^2 - 8x' - 36y' + 4 = 0$$

$$4(x' - 1)^2 + 9(y' - 2)^2 = 36$$

$$\frac{(x' - 1)^2}{3^2} + \frac{(y' - 2)^2}{2^2} = 1$$



Matrix representation:
Rewrite $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ using matrix products:

$$\underbrace{\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}}_{\text{quadratic}} + \underbrace{\begin{pmatrix} D & E \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + F}_{\text{linear}} = 0.$$

Let

$$\begin{pmatrix} x \\ y \end{pmatrix} = O \begin{pmatrix} x' \\ y' \end{pmatrix}, \qquad \text{where } O = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Note that O is orthogonal:

$$OO^T = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = O^T O.$$

Taking transpose:

$$\begin{pmatrix} x & y \end{pmatrix} = \begin{pmatrix} x' & y' \end{pmatrix} O^T$$

Put it into the quadratic part:

$$\begin{aligned} \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} x' & y' \end{pmatrix} O^T \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} O \begin{pmatrix} x' \\ y' \end{pmatrix} \\ &= \begin{pmatrix} x' & y' \end{pmatrix} \begin{pmatrix} A' & B'/2 \\ B'/2 & C' \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}. \end{aligned}$$

General Conics and Their Classification (Optional)

- $A + C = A' + C'$:

$$\begin{aligned} A + C &= \operatorname{tr}\begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} = \operatorname{tr}\left(O^T \begin{pmatrix} A' & B'/2 \\ B'/2 & C' \end{pmatrix} O\right) \\ &= \operatorname{tr}\left(\begin{pmatrix} A' & B'/2 \\ B'/2 & C' \end{pmatrix} OO^T\right) \\ &= \operatorname{tr}\begin{pmatrix} A' & B'/2 \\ B'/2 & C' \end{pmatrix} = A' + C' \end{aligned}$$

- $B^2 - 4AC = B'^2 - 4A'C'$:

$$\begin{aligned} AC - B^2/4 &= \det\begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} = \det\left(O^T \begin{pmatrix} A' & B'/2 \\ B'/2 & C' \end{pmatrix} O\right) \\ &= \det(O^T)\det\begin{pmatrix} A' & B'/2 \\ B'/2 & C' \end{pmatrix}\det(O) \\ &= \det\begin{pmatrix} A' & B'/2 \\ B'/2 & C' \end{pmatrix} = A'C' - B'^2/4 \end{aligned}$$

General Conics and Their Classification (Optional)

- Choosing a suitable θ such that $B' = 0$ corresponds to the orthogonal diagonalization of the symmetric matrix $\begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}$:

$$O^T \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} O = \begin{pmatrix} A' & 0 \\ 0 & C' \end{pmatrix}$$

The equation then becomes

$$\begin{pmatrix} x' & y' \end{pmatrix} \begin{pmatrix} A' & 0 \\ 0 & C' \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} + \begin{pmatrix} D & E \end{pmatrix} O \begin{pmatrix} x' \\ y' \end{pmatrix} + F = 0$$

$$A'x'^2 + C'y'^2 + \begin{pmatrix} D & E \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} + F = 0,$$

which represents

- an ellipse if $A'C' > 0$ (i.e. $B^2 - 4AC < 0$),
- a parabola if $A'C' = 0$ (i.e. $B^2 - 4AC = 0$),
- a hyperbola if $A'C' < 0$ (i.e. $B^2 - 4AC > 0$).

Equation of tangent

Example 11.10

Let $P(x_0, y_0)$ be a point on the conic section

$$\Gamma : Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

Show that the equation of tangent of Γ at P is given by

$$Ax_0x + B\frac{x_0y + y_0x}{2} + Cy_0y + D\frac{x + x_0}{2} + E\frac{y + y_0}{2} + F = 0.$$

By implicit differentiation,

$$2Ax + By + Bx\frac{dy}{dx} + 2Cy\frac{dy}{dx} + D + E\frac{dy}{dx} = 0.$$

Put $(x, y) = (x_0, y_0)$:

$$2Ax_0 + By_0 + D + (Bx_0 + 2Cy_0 + E)\frac{dy}{dx}\Big|_{(x_0,y_0)} = 0$$

Equation of tangent

So the equation of tangent of Γ at P is

$$2Ax_0 + By_0 + D + (Bx_0 + 2Cy_0 + E) \cdot \frac{y - y_0}{x - x_0} = 0$$

$$\begin{matrix} 2Ax_0x \\ -2Ax_0^2 \end{matrix} + \begin{matrix} By_0x \\ -Bx_0y_0 \end{matrix} + \begin{matrix} Dx \\ -Dx_0 \end{matrix} + \begin{matrix} Bx_0y \\ -Bx_0y_0 \end{matrix} + \begin{matrix} 2Cy_0y \\ -2Cxy0^2 \end{matrix} + \begin{matrix} Ey \\ -Ey_0 \end{matrix} = 0$$

$$\begin{matrix} 2Ax_0x + B(x_0y + y_0x) + 2Cy_0y + D(x + x_0) + E(y + y_0) \\ -2Ax_0^2 - 2Bx_0y_0 - 2Cy_0^2 - 2Dx_0 - 2Ey_0 \end{matrix} = 0$$

$$Ax_0x + B\frac{x_0y + y_0x}{2} + Cy_0y + D\frac{x + x_0}{2} + E\frac{y + y_0}{2} + F = 0.$$

Tangent of $\mathcal{P} : y^2 = 4ax$ at $P(x_0, y_0)$:

$$y_0y = 2a(x + x_0)$$

Tangent of $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P(x_0, y_0)$:

$$\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = 1$$

Tangent of $\mathcal{H} : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $P(x_0, y_0)$:

$$\frac{x_0x}{a^2} - \frac{y_0y}{b^2} = 1$$

Example 11.11 (Homework 10 Question 10)

Let $\mathcal{H} : xy = 1$, where $x, y > 0$, be a part of a hyperbola. Let $S = \left(s, \dfrac{1}{s}\right)$ and $T = \left(t, \dfrac{1}{t}\right)$ be distinct points on $\mathcal{H}$.

(a) Show that the equation of the chord ST is $x + sty = s + t$.

(b) Suppose that ST passes through $(2, 2)$.

  (i) Show that $t = \dfrac{s-2}{2s-1}$ and find all possible values of s.

  (ii) If M is the mid-point of ST, show that the equation of locus of M is $(x-1)(y-1) = 1$.

Example 11.11 (Homework 10 Question 10)

Let $\mathcal{H} : xy = 1$, where $x, y > 0$, be a part of a hyperbola. Let $S = \left(s, \frac{1}{s}\right)$ and $T = \left(t, \frac{1}{t}\right)$ be distinct points on $\mathcal{H}$.

(a) Show that the equation of the chord ST is $x + sty = s + t$.

(a) The equation of the chord ST is

$$\frac{y - \frac{1}{s}}{x - s} = \frac{\frac{1}{s} - \frac{1}{t}}{s - t}$$

$$\frac{y - \frac{1}{s}}{x - s} = -\frac{1}{st}$$

$$x + sty = s + t$$

Example 11.11 (Homework 10 Question 10)

Let $\mathcal{H} : xy = 1$, where $x, y > 0$, be a part of a hyperbola. Let $S = \left(s, \frac{1}{s}\right)$ and $T = \left(t, \frac{1}{t}\right)$ be distinct points on $\mathcal{H}$.

(a) Show that the equation of the chord ST is $x + sty = s + t$.

(b) Suppose that ST passes through $(2, 2)$.

(i) Show that $t = \dfrac{s-2}{2s-1}$ and find all possible values of s.

(b) (i) Put $x = 2$ and $y = 2$ into the equation obtained in (a),

$$2 + 2st = s + t$$

$$t = \frac{s-2}{2s-1}$$

Since $\dfrac{s-2}{2s-1} = t > 0$ and $s > 0$, we have $0 < s < \dfrac{1}{2}$ or $s > 2$.

Example 11.11 (Homework 10 Question 10)

Let $\mathcal{H} : xy = 1$, where $x, y > 0$, be a part of a hyperbola. Let $S = \left(s, \dfrac{1}{s}\right)$ and $T = \left(t, \dfrac{1}{t}\right)$ be distinct points on $\mathcal{H}$.

(b) Suppose that ST passes through $(2, 2)$.

(i) Show that $t = \dfrac{s-2}{2s-1}$ and find all possible values of s.

(ii) If M is the mid-point of ST, show that the equation of locus of M is $(x-1)(y-1) = 1$.

(b) (ii) Let $(x, y) = M = \left(\dfrac{s+t}{2}, \dfrac{\frac{1}{s}+\frac{1}{t}}{2}\right) = \left(\dfrac{s+t}{2}, \dfrac{s+t}{2st}\right)$.

From (b)(i), we have $2 + 2st = s + t$, so $x = 1 + st$ and $y = \dfrac{1+st}{st}$.

Put $st = x - 1$ into $y = \dfrac{1+st}{st}$,

$$y = \frac{1 + (x-1)}{(x-1)}$$

$$y(x-1) - (x-1) = 1$$

$$(x-1)(y-1) = 1$$

Example 11.12 (Homework 10 Question 11)

Let $P = (p, q)$ be a point which does not lie on the parabola $\mathcal{P} : y^2 = 4x$.

(a) If $y = mx + c$ is a straight line passing through the point P tangent to the parabola $\mathcal{P}$, show that $q = pm + \dfrac{1}{m}$.

(b) Given that the two tangents for the parabola $\mathcal{P}$ with slopes m_1 and m_2 intersects at the point P.

(i) Prove that $\left(\dfrac{1}{m_1} - \dfrac{1}{m_2}\right)^2 = q^2 - 4p$.

(ii) If $\left(\dfrac{1}{m_1} - \dfrac{1}{m_2}\right)^2 = 5$ and the point P lies on the line $y = x$, find the coordinates of P.

(c) If the tangents PT_1 and PT_2 are drawn to the parabola $\mathcal{P}$ where T_1 and T_2 are the points of contact. Let $\angle T_2PT_1 = \alpha$, where $\alpha \neq \dfrac{\pi}{2}$. Prove that the equation of the locus of the point P is

$$y^2 - 4x = (x+1)^2 \tan^2 \alpha.$$

What happens if $\alpha = \dfrac{\pi}{2}$?

Example 11.13 (Homework 10 Question 11)

Let $P = (p, q)$ be a point which does not lie on the parabola $\mathcal{P} : y^2 = 4x$.

(a) If $y = mx + c$ is a straight line passing through the point P tangent to the parabola $\mathcal{P}$, show that $q = pm + \dfrac{1}{m}$.

(a) Put $y = mx + c$ into the equation of $\mathcal{P}$,

$$\begin{aligned} (mx + c)^2 &= 4x \\ m^2x^2 + (2mc - 4)x + c^2 &= 0 \end{aligned}$$

Since the tangent line only intersects $\mathcal{P}$ at a single point, we have

$$\begin{aligned} \Delta &= 0 \\ (2mc - 4)^2 - 4(m^2)(c^2) &= 0 \\ -16mc + 16 &= 0 \\ c &= \frac{1}{m} \end{aligned}$$

Therefore, the tangent is $y = mx + \dfrac{1}{m}$. Furthermore, it passes through the point $P = (p, q)$. Therefore, $q = pm + \dfrac{1}{m}$.

Example 11.14 (Homework 10 Question 11)

Let $P = (p, q)$ be a point which does not lie on the parabola $\mathcal{P} : y^2 = 4x$.

(b) Given that the two tangents for the parabola $\mathcal{P}$ with slopes m_1 and m_2 intersects at the point P.

(i) Prove that $\left(\frac{1}{m_1} - \frac{1}{m_2}\right)^2 = q^2 - 4p$.

(ii) If $\left(\frac{1}{m_1} - \frac{1}{m_2}\right)^2 = 5$ and the point P lies on the line $y = x$, find the coordinates of P.

(b) (i) Rewriting the equation obtained in (a), we have $pm^2 - qm + 1 = 0$. Then, m_1 and m_2 are the roots of this equation, so $m_1 + m_2 = \frac{q}{p}$ and $m_1 m_2 = \frac{1}{p}$. Therefore,

$$\left(\frac{1}{m_1} - \frac{1}{m_2}\right)^2 = \frac{(m_1 - m_2)^2}{(m_1 m_2)^2} = \frac{(m_1 + m_2)^2 - 4m_1 m_2}{(m_1 m_2)^2}$$
$$= \frac{(\frac{q}{p})^2 - 4(\frac{1}{p})}{(\frac{1}{p})^2} = q^2 - 4p$$

(ii) We have $q^2 - 4p = \left(\frac{1}{m_1} - \frac{1}{m_2}\right)^2 = 5$ and $p = q$.

It implies $p = q = -1$ or $p = q = 5$, i.e. $P = (-1, -1)$ or $P = (5, 5)$.

Example 11.15 (Homework 10 Question 11)

Let $P = (p, q)$ be a point which does not lie on the parabola $\mathcal{P} : y^2 = 4x$.

(c) If the tangents PT_1 and PT_2 are drawn to the parabola $\mathcal{P}$ where T_1 and T_2 are the points of contact. Let $\angle T_2PT_1 = \alpha$, where $\alpha \neq \frac{\pi}{2}$. Prove that the equation of the locus of the point P is

$$y^2 - 4x = (x+1)^2 \tan^2 \alpha.$$

What happens if $\alpha = \frac{\pi}{2}$?

(c)

$$\begin{aligned}
|\tan \alpha| &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\
\tan^2 \alpha &= \frac{(m_1 - m_2)^2}{(1 + m_1 m_2)^2} = \frac{(m_1 + m_2)^2 - 4m_1 m_2}{(1 + m_1 m_2)^2} \\
&= \frac{(\frac{q}{p})^2 - 4(\frac{1}{p})}{(1 + \frac{1}{p})^2} = \frac{q^2 - 4p}{(p+1)^2} \\
q^2 - 4p &= (p+1)^2 \tan^2 \alpha
\end{aligned}$$

which gives an equation in p and q. The result follows by replacing p and q with x and y.

Example 11.16 (Homework 10 Question 11)

Let $P = (p, q)$ be a point which does not lie on the parabola $\mathcal{P} : y^2 = 4x$.

(c) If the tangents PT_1 and PT_2 are drawn to the parabola $\mathcal{P}$ where T_1 and T_2 are the points of contact. Let $\angle T_2PT_1 = \alpha$, where $\alpha \neq \frac{\pi}{2}$. Prove that the equation of the locus of the point P is

$$y^2 - 4x = (x+1)^2 \tan^2 \alpha.$$

What happens if $\alpha = \frac{\pi}{2}$?

(c) When $\alpha = \frac{\pi}{2}$, we have $\frac{1}{p} = m_1 m_2 = -1$, i.e. $p = -1$. Therefore, P moves along the vertical line $x = -1$, which is exactly the directrix of $\mathcal{P}$.

Example 11.17 (HKALE Pure Math 1996 Paper 2 Q11)

Figure 1 shows a circle with radius b rolling externally without slipping on a fixed circle with radius a and centred at the origin. Let P be a point fixed on the rolling circle with initial position at $A(a, 0)$.

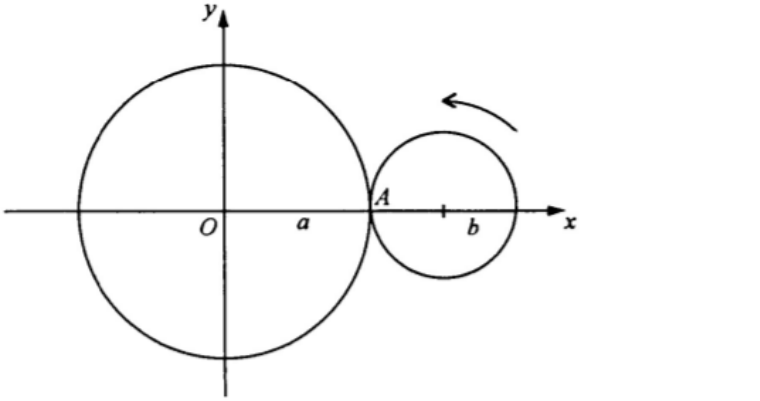


Figure 1

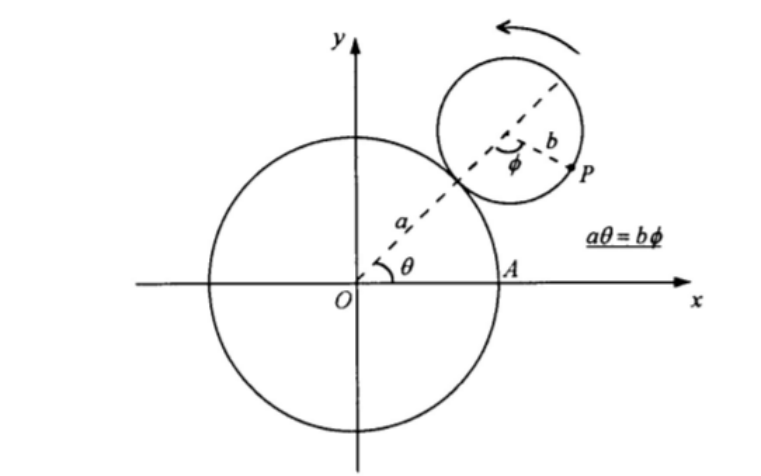


Figure 2

(a) Referring to Figure 2, show that the parametric equations of the locus of P are given by

$$\begin{cases} x = (a+b)\cos\theta - b\cos\left(\dfrac{a+b}{b}\right)\theta \\ y = (a+b)\sin\theta - b\sin\left(\dfrac{a+b}{b}\right)\theta \end{cases}.$$

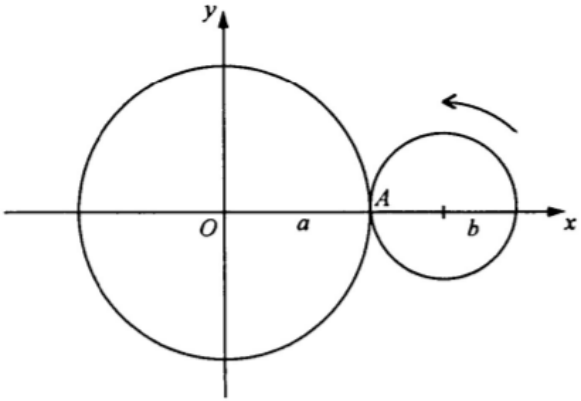


Figure 1

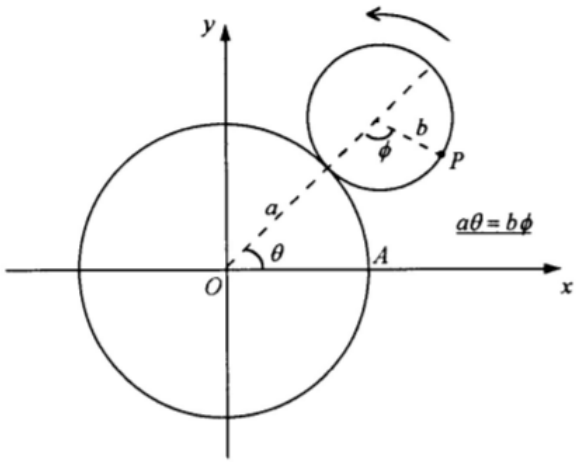


Figure 2

Let C be the center of the rolling circle. Then

$$\begin{aligned}\overrightarrow{OC} &= (a+b)(\cos\theta, \sin\theta),\\ \overrightarrow{CP} &= b(\cos(-(\pi-\theta-\phi)), \sin(-(\pi-\theta-\phi)))\\ &= b(-\cos(\theta+\phi), -\sin(\theta+\phi)).\end{aligned}$$

Since $a\theta = b\phi$, we have

$$\overrightarrow{OP} = \overrightarrow{OC} + \overrightarrow{CP} = ((a+b)\cos\theta, (a+b)\sin\theta) + (-b\cos(\theta + \frac{a\theta}{b}), -b\sin(\theta + \frac{a\theta}{b}))$$

So the parametric equations of the locus of P are given by

$$\begin{cases} x &= (a+b)\cos\theta - b\cos\left(\dfrac{a+b}{b}\right)\theta \\ y &= (a+b)\sin\theta - b\sin\left(\dfrac{a+b}{b}\right)\theta \end{cases}.$$

Example 11.18 (HKALE Pure Math 1996 Paper 2 Q11)

Figure 1 shows a circle with radius b rolling externally without slipping on a fixed circle with radius a and centred at the origin. Let P be a point fixed on the rolling circle with initial position at $A(a, 0)$.

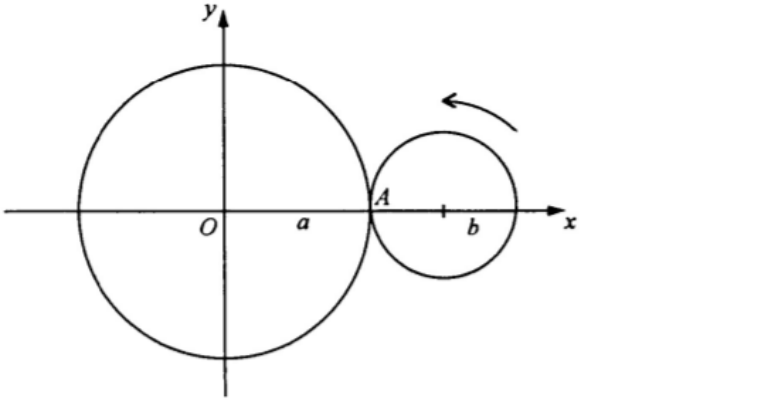


Figure 1

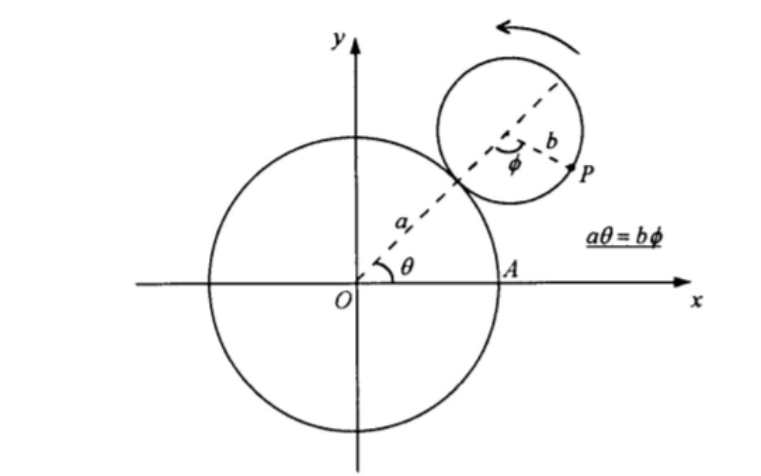


Figure 2

(b) Suppose $a = 2b$

(i) Write down the parametric equations of the locus of P.

(ii) Express $\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}$ in terms of θ.

(iii) Find the distance travelled by P before it meets the fixed circle again.

Suppose $a = 2b$.

(i) The parametric equations of the locus of P:

$$\begin{cases} x &= 3b\cos\theta &- b\cos 3\theta \\ y &= 3b\sin\theta &- b\sin 3\theta \end{cases}.$$

(ii)

$$\begin{cases} \dfrac{dx}{d\theta} &= -3b\sin\theta &+ 3b\sin 3\theta \\ \dfrac{dy}{d\theta} &= 3b\cos\theta &- 3b\cos 3\theta \end{cases}.$$

$$\begin{aligned} \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= 9b^2(1 + 1 - 2\sin\theta\sin 3\theta - 2\cos\theta\cos 3\theta) \\ &= 9b^2(2 - 2\cos 2\theta) = 9b^2(2 - 2(1 - 2\sin^2\theta)) \\ &= 36b^2\sin^2\theta \end{aligned}$$

$$\therefore \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = 6b|\sin\theta|.$$

(iii) When P first meet the fixed circle again, $\phi = 2\pi$. So $\theta = \frac{b}{a}\phi = \pi$.

$$\text{Distance travelled by } P = \int_0^\pi 6b^2|\sin\theta|\,d\theta = 12b.$$

Example 11.19

Let P, Q be points moving on an ellipse $\mathcal{E} : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ such that $\angle POQ = \frac{\pi}{2}$ and the tangents at P, Q intersect at T. Find the equation of locus of T.

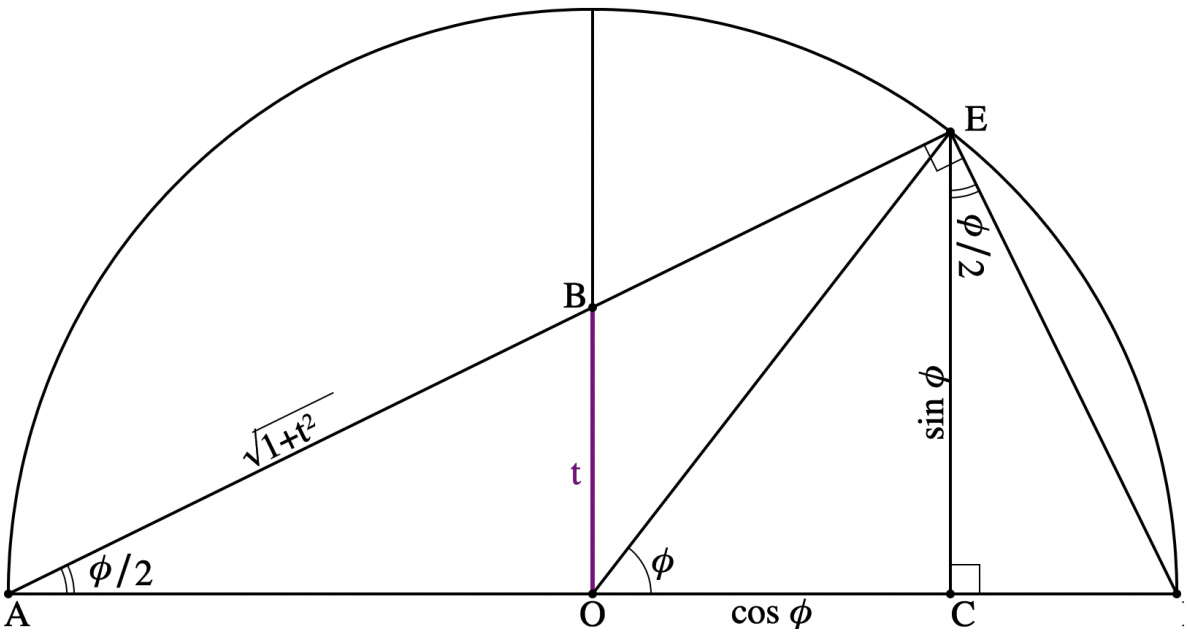
Equations of tangents at $P = (x_1, y_1), Q = (x_2, y_2)$:

$$\frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = 1$$
$$\frac{x_2 x}{a^2} + \frac{y_2 y}{b^2} = 1.$$

Solving the equations, $T = (x, y)$ is given by

$$x = \frac{-a^2(y_2 - y_1)}{x_2 y_1 - x_1 y_2} \qquad y = \frac{b^2(x_2 - x_1)}{x_2 y_1 - x_1 y_2}.$$

Consider the stereographic projection of the unit circle onto the *y-axis*:



- $\tan\frac{\phi}{2} = t$
- $\sin\phi = \frac{2t}{1+t^2}$
- $\cos\phi = \frac{1-t^2}{1+t^2}$
- $\tan\phi = \frac{2t}{1-t^2}$

A parametrization of the unit circle (except the point $(-1,0)$) is

$$\left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2}\right), \quad t \in \mathbb{R}.$$

Hence, we have a parametrization of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

$$\gamma(t) = \left(a \cdot \frac{1-t^2}{1+t^2}, b \cdot \frac{2t}{1+t^2}\right), \quad t \in \mathbb{R}.$$

Let $P = \gamma(t_1)$, $Q = \gamma(t_2)$. Then

$$x_2 - x_1 = a\frac{(1-t_2^2)(1+t_1^2) - (1-t_1^2)(1+t_2^2)}{(1+t_1^2)(1+t_2^2)} = \frac{-2a(t_2^2 - t_1^2)}{(1+t_1^2)(1+t_2^2)}$$

$$y_2 - y_1 = 2b\frac{(t_1)(1+t_2^2) - (t_2)(1+t_1^2)}{(1+t_1^2)(1+t_2^2)} = \frac{2b(t_2 - t_1)(1 - t_1t_2)}{(1+t_1^2)(1+t_2^2)}$$

$$x_2y_1 - x_1y_2 = ab\frac{(1+t_2^2)(2t_1) - (1+t_1^2)(2t_2)}{(1+t_1^2)(1+t_2^2)} = \frac{-2ab(t_2 - t_1)(1 + t_1t_2)}{(1+t_1^2)(1+t_2^2)}.$$

So

$$x = \frac{-a^2(y_2 - y_1)}{x_2y_1 - x_1y_2} = \frac{a(1 - t_1t_2)}{1 + t_1t_2}$$

$$y = \frac{b^2(x_2 - x_1)}{x_2y_1 - x_1y_2} = \frac{b(t_1 + t_2)}{1 + t_1t_2}.$$

Now $\angle POQ = \frac{\pi}{2}$ means that

$$\frac{2bt_1}{a(1-t_1^2)} \cdot \frac{2bt_2}{a(1-t_2^2)} = -1$$

$$4b^2t_1t_2 = -a^2(1-t_1^2)(1-t_2^2).$$

Want to express this relation in terms of $(1 - t_1t_2)$, $(t_1 + t_2)$ and $(1 + t_1t_2)$.

Note

$$4b^2t_1t_2 = b^2[(1+t_1t_2)^2 - (1-t_1t_2)^2],$$

and

$$\begin{aligned}
-a^2(1-t_1^2)(1-t_2^2) &= -a^2(1-t_1)(1-t_2)(1+t_1)(1+t_2) \\
&= -a^2(1+t_1t_2-t_1-t_2)(1+t_1t_2+t_1t_2) \\
&= -a^2[(1+t_1t_2)^2-(t_1+t_2)^2].
\end{aligned}$$

The equation $4b^2t_1t_2 = -a^2(1-t_1^2)(1-t_2^2)$ becomes

$$(a^2+b^2)(1+t_1t_2)^2 = b^2(1-t_1t_2)^2 + a^2(t_1+t_2)^2.$$

Substituting $x = \dfrac{a(1-t_1t_2)}{1+t_1t_2}$, $y = \dfrac{b(t_1+t_2)}{1+t_1t_2}$ back into the formula, we have

$$\begin{aligned}
a^2+b^2 &= b^2(\frac{x}{a})^2 + a^2(\frac{y}{b})^2 \\
\frac{x^2}{a^4} + \frac{y^2}{b^4} &= \frac{1}{a^2} + \frac{1}{b^2}.
\end{aligned}$$